

**EVALUATION OF LAND-USE AND LAND-COVER CHANGES IN AWORO
FOREST RESERVE, OGUN STATE, NIGERIA**

BY

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DECLARATION

I hereby declare that this project was carried out by me, and has not been presented for a degree or other awards in any college or institution in part, or in full.

CERTIFICATION

This is to certify that this study was carried out by OYENEYE, Sarah Arinola, 18/10AS078 in the Department of Forest Resources Management, Faculty of Agriculture, University of Ilorin. This project has been read and approved in partial fulfillment of the requirements for the award of Bachelor of Forestry and Wildlife (B. Forestry and Wildlife) Degree in the University.

DEDICATION

This project is dedicated to God Almighty, who has made it a possibility, and to my family, most especially my parents, for their love and encouragement. I pray that God, in His infinite mercy, will grant you long life to eat the fruit of your labour, and also to everyone that has contributed to the success of this work. I say thank you and God bless you all.

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ABSTRACT

This study was conducted to investigate the land-use and land-cover situation and serial changes in Aworo Forest Reserve, Ogun State in relation to its management. A combination of remote sensing and GIS techniques was utilized to evaluate changes that occurred between 1984 and 2023 in the study area. Landsat images of 1984 (TM), 2000 (ETM), 2013 (OLI/TC), 2020 (OLI/TC) and 2023 (OLI/TC) were used to evaluate the spatial extent, change trends and dynamics within the forest reserve. Field observation (ground-truthing) was carried out in the area in order to validate pre-classified images of the study area and for appropriate classifications. The acquired images alongside the ground-truth data were analyzed using ArcGIS 10.8. Unsupervised (Iso-cluster) classification method was used. Image classification accuracies were assessed using Kappa coefficient and other accuracy statistics, including confusion matrix. The result of image analysis produced five major land-uses/covers classes, including: primary forest, secondary forest, agricultural land, fallow land and bareland/built-up area. The results showed that between 1984 and 2023, primary forest declined by 1,886 ha (-40.0%) at about 1.2% per annum. In the same vein, 4,502 ha (56.8%) of fallow land was lost, at a rate of 1.5% per annum. However, secondary forest, agricultural land and bareland/built-up increased by 441 ha (8.1%), 5,146 ha (87.6%) and 801 ha (58.7%), respectively, at annual rates of 0.2%, 2.3%, and 1.5%, respectively. The major drivers of forest-cover losses were intensive agriculture practices and over-exploitation of timber. The results further showed a decreasing trend for forest-cover in the future while agricultural land and bareland/built-up were predicted to be on the rise, and may lead to significant degradation, should the trend be allowed to subsist. Therefore, better land-use management and environmental-conservation interventions are advocated in the study area.

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CHAPTER ONE

1.0 INTRODUCTION

1.1 Background to the Study

According to Ladan (2015), forests are important plant communities that consist of trees and other woody vegetation that performs life supporting functions on earth. Governments, especially in semi-arid regions of the world, designate areas of forest as forest reserves to stimulate rainfall, reduce wind erosion, stem the tide of desertification and halt the encroachment of the desert. Therefore, in many countries, forest reserves enjoy judicial and/or constitutional protection under a legal system.

Forests are the next most important resources of nature on earth after air and water. They essentially support life on earth by absorbing carbon dioxide and releasing oxygen, thereby maintaining balance in the gaseous atmosphere and also in completion of hydrological cycle to cause rainfall. It is based on the above importance of forests that the United Nations suggested that 25% of the surface area of every country should be conserved under permanent forest cover, as the minimum ecological requirement for the socio-economic survival of the country and its people (Bugaje, 2007).

According to Soaga *et al.* (2012), tropical countries in Africa, especially Nigeria, lay emphasis on management and protection of forest through forest reservation. This process of reservation has significant social, economic and environmental impacts.

Reservation of forests, especially urban and peri-urban forests, gives rise to benefits in terms of conservation of biodiversity. The biodiversity of plants and animals provides goods and services. The myriad of goods and services provided by urban and peri-urban forests have been widely reported (FAO, 2001), they are timber and non-timber based, examples include sawn wood, fuel wood, medicines, leaves, roots, bark, mushroom, wild animals and geological materials such as sand, gravels, stones and soils. Consequently, the forest is a key component and a natural resources base of any community or country especially in developing countries (Soaga *et al.*, 2012).

Land-use and land-cover changes are worldwide ecological processes, which result in major environmental challenges of global importance (Sleeter *et al.*, 2012; Klimanova *et al.*, 2018). Land-use/cover change is arguably the most pervasive socio-economic force driving change and degradation of ecosystem. Deforestation, urban development, agriculture and other human activities have substantially altered the earth's landscape (Wu, 2008). Nowadays, agriculture occupies 38% of the globe's ice-free terrestrial surface, and is the largest land-cover type by area (FAO, 2011). The rate of agricultural expansion is increasing globally (Abeselom, 2018).

Many researchers argue that LULCC emerged as a major aspect in the wider debate of global change, and that change originates from human-induced impacts on the environment and their implications for climate change (Lambin and Geist, 2001; Woldeamlak, 2002; Ginblett, 2006). The indicators of these changes can be clearly seen in the current major global concerns such as increasing concentrations of carbon dioxide

(CO₂) in the atmosphere, loss of biological diversity, conversion and fragmentation of natural vegetation areas as well as accelerated emission of greenhouse gases (Steffen, 2001).

As one of the states with forest resources, Ogun State has forest reserves that fall under the purview of peri-urban forest and some in rural areas (Soaga *et al.*, 2012). Peri-urban was described as areas beyond the closed settled limits of any urban area but is sufficiently close enough to the urban area to have frequent and substantial interactions with the urban economy (Wiggins and Holt, 2000).

1.2 Statement of Problem

According to Adebayo *et al.* (2019), activities of man on earth surface and some natural disasters have greatly revolutionized, and negatively impacted the natural environment altering the form of natural land covers. Studies have also shown that only a few landscapes are still in their natural state globally (Manish and Aruna, 2011). Natural phenomena such as storms, wildfires, pest, earthquake, draught and the likes have also altered the form of natural land covers (Jwan *et al.*, 2013). This had resulted in significant pattern in the changes of land use/land cover (Adebayo *et al.*, 2019). Information on land-use situation in Aworo Forest Reserve appears to be non-existent.

The natural environment particularly in Nigeria is gradually going into extinction at a rapid rate due to increasing population, which is accompanied by demand for resources. The need for more habitats, places to live and protect humans from the various

factors of environment has led to the destruction of natural resources. Due to serious anthropogenic activities, the earth surface is being significantly altered in several ways. Forest and vegetation covers are getting depleted day by day at a greater rate (Mengistu and Salami, 2007). Nevertheless, there is dearth of information on land use and land cover in Aworo Forest Reserve. Again, the possible drivers of change in the reserve are still unknown.

The growing population and increasing socio-economic necessities may have created a pressure on land use and land cover in the forest reserve, but the extent of damages these may have caused have not been documented. The land-use and land-cover alterations are generally believed to have been caused by mismanagement of agricultural, urban, range and forest lands, however, this has not been substantiated by empirical data from the study area. Although, few similar studies exist within other state's forest reserves (e.g., Adeyemi and Adeleke, 2020; Adeyemi and Oyeleye, 2021; Adeyemi and Ayinde, 2022; Adeyemi and Olowo, 2022), the current dimension in forest-cover change of Aworo Forest Reserve is yet unknown.

1.3 Objectives of the Study

The main objective of this study is to assess the forest-cover condition in Aworo Forest Reserve over the years. The specific objectives are to:

- (i) investigate the current land-use pattern in the study area;

- (ii) examine the land-use and land-cover changes in the area in the last five decades;
- (iii) investigate the drivers of the forest-cover change dynamics in the study area;
- (iv) simulate future scenarios, if the prevailing conditions persist.

1.4 Justification for the Study

Land-cover changes occur naturally, in a progressive and gradual way, however, it may sometimes be rapid and sudden due to anthropogenic activities (Butenuth *et al.*, 2007). Thus, analyzing the land-cover changes, and understanding the subsequent trends of changes contribute to present complex dynamics of land cover, and are important for policy making, planning and implementation of natural resource management (Ioannis and Meliadis, 2011; Knorr *et al.*, 2011; Reddy and Gebreselassie, 2011).

According to Igbokwe (2010), land-use information should form part of the environmental data, which are kept in the form of inventories/infrastructures in an emerging economy. Remote Sensing and Geographic Information System (GIS) are advanced methods in capturing, digitizing, analyzing, processing and interpreting land-use/land-cover data. Nowadays, integration of GIS and remote sensing has provided accurate information about land-use changes (Imam, 2011; Alqurashi and Kumar, 2013) for informed decision makings.

CHAPTER TWO

2.0

LITERATURE REVIEW

2.1 Forest Reserve

Reserves allow people to experience and understand how forest ecosystems function when timber and other wood products that are normally extracted for human use remain in place. While it is important to have the great majority of forestland open to the sustainable harvest of wood products in order to support human society, it is equally important to retain portions of the forested landscapes in a condition where all components of the ecosystem remain in place.

Forest reserves are therefore portions of vegetated state lands that are dominated by trees and other woody shrubs, where commercial harvesting of wood products is excluded in order to capture elements of biodiversity that can be missing from sustainably harvested sites (Commonwealth of Massachusetts, 2011). However, in developing countries like Ghana, portions of forest reserves are sometimes given to timber concessionaires for commercial harvesting. Savanna forest reserves are thus Savanna woodland vegetated areas that are demarcated and protected for productive and protective functions. The importance of forest reserves in the lives of humanity cannot be underestimated. For instance, forest reserves in many African countries are sources of non-wood forest products which play a crucial role in the welfare of the local population.

In Nigeria, forests reserves provide livelihood for people living in and around them and serve as vital safety nets for the rural poor, however, forest resources are being depleted at alarming rates (Mbaya and Hashidu, 2017). Therefore, Man's incursions into the original forest for the purpose of development has resulted in biodiversity loss thereby increasing reduction in the available forest reserves, loss and extinction of flora and fauna with a consequent decline in food with an expected long-term effect from climate change (Saha and Guru, 2003). The State Forestry Departments in Nigeria are the most important institutions in forest management in the country because the constitution grants them the sole right to manage the country's forest resources, not on behalf of the country, but rather, on behalf of their specific states (World Bank, 2005). However, some salient concerns, ranging from 'ownership issue' through 'definition of roles' to 'management right and tenure,' which have been left largely unresolved among the federating units (Federal, State and Local government) since the commencement of democracy in 1999, constitute serious impediments toward tackling large scale biodiversity loss (Mbaya and Hashidu, 2017).

2.2 Land Use and Land Cover

Land cover refers to the physical characteristics of earth's surface, captured in the distribution of vegetation, water, soil and other physical features. Land cover also refers to the actual surface cover for a given location (e.g., vegetation type, anthropogenic structure, etc.). Remote-sensing data have a long history of being used for deriving land-

cover maps, even before the launch of the first Landsat platform in 1972 (Sohl and Sleeter, 2012). Aerial photography served as a primary source of information on land cover before the availability of satellite imagery, and it remains an important source of land-cover information even today (Akbari *et al.*, 2003). Aerial photography, which was available before the launch of the first Landsat, remains a valuable tool for analyzing historical LULC change (Thomson *et al.*, 2007; Gerard *et al.*, 2010). With the advent of Landsat and other commercial remote-sensing satellites, land-cover mapping at all scales has flourished. Land-cover information at multiple spatial, thematic, and temporal resolutions have direct relevance to LULC forecast modeling (Sohl and Sleeter, 2012).

Land use refers to the way in which land has been used by humans and their habitats (such as agriculture, settlements, industry etc.). Unlike land cover, which can be directly observed and monitored from remote-sensing data, land use typically must be inferred through a combination of remote-sensing observation, regional and local knowledge (including field observation), and other ancillary information that links a given land cover in a region with a given land use (Sohl and Sleeter, 2012). Brown *et al.* (2008) used a combination of MODIS imagery, land-cover classification, information from the 2001 NLCD (Homer *et al.*, 2007), and county-level irrigation statistics to create a land-use map of irrigated cropland.

Although land use is generally inferred based on the cover, yet both the terms land use and land cover being closely related are interchangeable. For example,

settlement is cover but if we include buildings whether it is being used for residence or industrial activity, it shows the land-use component (Chaudhary *et al.*, 2008).

2.2.1 Land-use/Land-cover Changes

Land-use and land-cover change is a general term for human modification of earth's surface. Though humans have been modifying land to obtain food and other essentials for thousands of years, current rates, extents and intensities of land-use and land-cover changes are far greater than ever in history, driving unprecedented changes in ecosystems and environment processes at local, regional and global scales. These changes encompass the greatest environmental concerns of human populations today, including climate change, biodiversity loss and the pollution of water, soils and air. Monitoring and mitigating the negative consequences of land-use and land-cover changes while sustaining the production of essential resources has therefore become a major priority of researchers around the world. Land-use and land-cover changes have become a central component in current strategies for managing natural resources and monitoring environmental changes. Since environmental changes are ubiquitous throughout the world, analyzing symptoms of emerging environmental change at local levels are essential (Iwuji *et al.*, 2017).

Land cover includes earth's land surface distribution of vegetation, water, desert and ice as well as the biota, soil, topography etc. in the immediate subsurface, and it also includes human activity areas, such as settlement, mine exposure etc. (Lambin *et al.*,

2003; Oumer, 2009). On the other hand, land use is attributed to how humans exploit the land cover to serve their own purposes and includes features such as residential zones, agricultural farms, logging areas etc. (Zubair, 2006; Oumer, 2009). In this context, land use influences the changes in land cover; therefore, LULC change can be defined as the modification of surface features on earth's landscape which is realized by the difference in their surface appearance assessed at two different times (Ayele, 2011). The current global condition of mass environmental change and sustainability issues elucidates the gravity of LULC change detection research in different parts of the world. Though LULC changes entail natural (e.g. weather, flooding, earthquake etc.) and anthropogenic causes, the ever-increasing demand of the mushrooming population has designated the anthropogenic influences as the most dramatic (Turner *et al.*, 2007; Foley *et al.*, 2011; Weinzettel *et al.*, 2013). At present, the undisturbed pristine areas cover less than 50% of the total earth's landscape; forest cover is only 30% which was around 50% some 8000 years ago (Oumer, 2009).

2.2.2 Causes of Land-use/Land-cover Change

Land-use/land-cover changes reflect the dynamics observed in the socio-economic condition of a given area. Similarly, changes in the socio-economic situations causes land-use/land-cover changes through their influence on land management techniques used and other various aspects of the farming systems, institutional settings, environmental policy and others. As many researchers indicated in Nigeria, several

factors have been modifying the original form of land cover. These include human activities such as agricultural colonization, spread of rural settlements, evolution of rural road networks and government policy (Salami, 2001).

According to Meludu (2004), the forest sub-sector accounts for about 8% of the agricultural gross domestic product of the country. He also noted that the estimated benefit of biodiversity in general (in billion dollars), by the Federal Environmental Protection Agency in Nigeria amounts to \$03.75 GDP. Current benefits (various species, subsistence, ecosystem, scientific and recreational) are \$06.00; non-current, option, existence and inheritance uses are \$01.00 and direct employment benefits are \$15.60. All these will amount to about \$26.35 billion. Despite these numbers, biodiversity depletion continues unchecked. Therefore, socio-economic survey is important to identify the factors that drive changes in land use/land cover of an area. The socio-economy of Nigeria in general and the south western part in particular is changing based on the national goal of transforming the country from oil based to agricultural-based economy during the past two decades (that is since 1991). The change in the economic system of the region puts pressure on the agricultural sector. This in turn leads to the opening up of more natural forests for cultivation. Other important factors of the land-use/land-cover change are the expansion of settlement and illegal logging in the area.

According to the report of Salami (2001), the introduction of exotic species plantations into the forest reserves that were originally designed to serve as repositories of the natural species of the forest belt has modified the ecological setting of the region.

The process involves extensive clearing and burning of the native species (original species) before planting exotics. All these have ecological implications in the area. In addition to the above noted human factors, climate change has significant impact on land-use/land-cover changes observed during the study period (Mengistu and Salami, 2007). In this regard, Fasona and Omojola (2005) report rainfall changes, which is a critical element of climate and climatic changes in tropical Africa, in the guinea-sudan-sahel (GSS) zone of Nigeria over a six decade (1940 - 2000 period). According to their report, the decade 1950s recorded the highest rainfall while the decade 1980s had the least rainfall from the total decadal mean (Mengistu and Salami, 2007). The pattern of land-cover changes between 1976 and 1995 strongly indicated loss of prime arable lands resulting from climate change, which is in turn leading to opening up of new virgin lands towards the south (Fasona and Omojola, 2005). These are strong signals of the effects of climate change on land-use/land-cover change at local level. Generally, the increasing down-south march of the Sahara Desert through the Sahelian zone of Nigeria is leading to the opening up of more natural forests for cultivation and grazing in the guinea savanna and rain forest zones (Mengistu and Salami, 2007).

2.2.3 Consequences of Land-use/Land-cover Changes

Tropical deforestation has been an environmental concern (Azeez, 2002) affecting both the forest areas and forest communities. Land-use/land-cover change has far reaching ecological consequences. The continuous degradation of the forest has major

effects on other segments of the environment. The manifestations of these have been noted to be in form of the rapid disappearance of forest cover leading to erosion, loss of biodiversity, soil degradation, and unfavorable hydrological changes. Regarding the soil degradation, the replacement of natural forest by plantations of exotics has negative effects on the soil. Once the tropical rain forest is disturbed, the organic matter content of the soil declines until restoration of a cropping system or forest that contributes carbon to the soil as rapidly as it is lost through oxidation. However, the high population density has led to a situation in which the land cleared from forest, is re-cultivated at repeated intervals and not allowed to regenerate into mature secondary forest (Mengistu and Salami, 2007).

According to Olaniran (2002), in Nigeria, deforestation has been seen to cause erosion that is affecting reservoirs, decreasing the aquatic populations, reducing the navigability of canals and also the quality of soil. He also noted that the problem of desertification and environmental degradation caused by land-use/land-cover change threatens the country (Mengistu and Salami, 2007).

Land use/land cover change also has an adverse effect on the biodiversity. In this regard, Ojo (2004), in his study of Abeku sector of Omo forest reserve indicated that there were 98, 109, 95 and 71 plant species enumerated in 1985, 1987, 1997 and 2000 respectively. According to his report, the increase from 1985 to 1987 was due mainly to more species being identified in the 1987 enumeration but the subsequent decrease between 1997 and 2000 was due largely to long interval between measurement during

which the plots had been encroached and some plots completely lost to cash and arable crop farming. This affected the diversity of the tree species in the forest reserve (Mengistu and Salami, 2007).

Over half of the world's landscape is influenced by human activities or under some sort of anthropogenic development and since the historic past, many natural resources have been heavily used or even depleted in the worst cases (Foley *et al.*, 2005). The impacts of this widespread LULC change on the natural environment are multifaceted, including climate change, alteration of hydrological cycle, increased water extraction, impairment of water quality, degradation of soil nutrients, amplified surface erosion, and loss of biodiversity (Turner *et al.*, 2007). Therefore, information on land-use and land-cover, changing trends and optimal use of the land resources have become predestined criteria for land-use planning and effective natural resources management of an area.

Generally, it has been noted that land degradation is both a part and consequence of environmental changes leading to loss of valuable land resources. There is the need therefore to strike equilibrium in the uses to which the land is put in order to minimize the far reaching ecological consequences of land-use/land- cover change (Mengistu and Salami, 2007).

2.2.4 Land-use and Land-cover Dynamics

Land-use/land-cover dynamics are widespread, accelerating, and significant processes majorly impelled by human actions and at the same time resulting to changes that impact human livelihood (Agarwal *et al.*, 2002). The LULC dynamics modify the availability of different important resources including vegetation, soil, water, and others (Bruijnzeel, 2004).

The modification of earth's terrestrial surface by human activities is commonly known as Land-use/land-cover change (LULCC) around the globe. Although modification of land by humans to obtain livelihoods and other essentials has been there for thousands of years, the extent, intensity and rate of LULCC are far greater now than it was in the past. These changes are driving forces for local, regional and global level unprecedented changes in ecosystems and environmental processes. Thus, LULC changes play an important role in the study and analysis of global changed scenario today as the data available on such changes is essential for providing critical input to decision making of ecological management and environmental planning for future (Zhao *et al.*, 2004; Dwivedi *et al.*, 2005; Fan *et al.*, 2007).

According to Rahdary *et al.* (2008), LULC change is dynamic, widespread and accelerating process, mainly driven by natural phenomena and anthropogenic activities. Geist and Lambin (2002) also stated constant changes in response to the dynamic interaction between underlying drivers and proximate causes. In general, LULC changes

are affected by human-induced activities and growth, socio-economic factors, deterioration of vegetation cover, agricultural activities, government policies and environmental factors (Gol *et al.*, 2010). Only few landscapes remain on the earth that are still in their intact natural state, the major causes of LULC dynamism being primarily associated with agricultural activities. Crop land and pastures are now among the dominant ecosystems on the planet, occupying more than 35% of the world's ice-free land surface (Paul and Lisa, 2011). Empirical studies by researchers from diverse disciplines found that changes in land use/land cover has become key to many diverse applications such as agriculture, environment, ecology, forestry, geology and hydrology (Weng, 2001).

2.2.5 Land-cover Classification

According to Aplin (2004), land cover is an intrinsic element of most remote sensing analysis. An obvious example is the use of remotely sensed imagery for straightforward land-cover classification (Franklin and Wulder, 2002; Alvarez *et al.*, 2003). Less obvious, perhaps, is the role of land cover in disease mapping (Tran *et al.*, 2002). Aplin (2004), the reason land cover has such a key position in the field of remote sensing relates to the way remotely sensed data are acquired. Land-based features may be categorized according to land-cover classes (e.g., grass, concrete, water, etc.) (Smith *et al.*, 2003), and since different land-cover features reflect radiation in different ways, remotely sensed images provide a basic representation of land-cover variation on the

earth's surface. Even where land-cover information is not the ultimate goal of remote sensing studies, it is often a useful aid for further analysis (Aplin, 2004).

Perhaps the most basic form of land-cover analysis within the field of remote sensing is land-cover classification. This involves the association of features within remotely sensed imagery (often, pixels) with specific land-cover classes and results in the production of land-cover maps (Smith *et al.*, 2003). Aplin (2004), land-cover classification has been used widely for over two decades (Briem *et al.*, 2002; King, 2002), yet there continues to be much interest in developing new classification techniques or adapting old ones (Ju *et al.*, 2003; Peng *et al.*, 2003). In particular, research efforts are being made to refine sub-pixel classification techniques, the means by which individual pixels are assigned proportional membership to multiple land-cover classes rather than full membership to a single class (Hagen *et al.*, 2002; Huang and Townshend, 2003). An adapted sub-pixel classifier was presented by (Ju *et al.*, 2003), that combine the structural simplicity of a linear mixture model and the discriminatory power of a neural network (Aplin, 2004).

Another area of considerable recent interest is in the development of multiple classifiers, the process of combining independent classification algorithms to increase the accuracy with which land-cover maps are produced (Debeir *et al.*, 2002; Liu *et al.*, 2002). Bruzzone and Cossu (2002) combined both multiple classification algorithms and multi-temporal imagery in a novel cascading classification structure to enable the regular updating of land-cover maps. In addition to work on developing classification techniques,

there has been much recent interest in testing different data sources for land-cover classification (Briem *et al.*, 2002; Liu *et al.*, 2003).

Other studies have merged contextual or other ancillary data with remotely sensed imagery to increase the accuracy of land-cover classification (Aplin, 2004). Liu *et al.* (2003), for instance, introduced geophysical information such as elevation and temperature into the classification procedure, and Debeir *et al.* (2002) included topographic features such as roads and rivers.

2.3 Effects of Land-use/Land-cover Changes at Local, Regional and Global Scale

2.3.1 Climate Change

In terms of climate change, since 1850, about 35% of the human caused emissions of CO₂ to the atmosphere comes from land-use changes (Foley *et al.*, 2005), and about 38% of earth's land area has been converted to agriculture (Foley *et al.*, 2011).

Human activities have modified the environment for thousands of years. Significant population increase, migration, and accelerated socio-economic activities have intensified these environmental changes over the last several centuries. The climate impacts of these changes have been found in local, regional, and global trends in modern atmospheric temperature records and other relevant climatic indicators. An important human influence on atmospheric temperature trends is extensive land-use/land-cover change and its climate forcing (Mahmood *et al.*, 2010). Studies using both modeled and observed data have documented these impacts (e.g., Kalnay and Cai, 2003; Cai and

Kalnay, 2004; Trenberth, 2004; Vose *et al.*, 2004; Feddema *et al.*, 2005; Christy *et al.*, 2006; Ezber *et al.*, 2007). Thus, it is essential that we detect LULCCs accurately, at appropriate scales, and in a timely manner so as to better understand their impacts on climate and provide improved prediction of future climate (Mahmood *et al.*, 2010). NRC (2005) has recommended the broadening of the climate-change issue to include LULCC processes as an important climate forcing. Mahmood *et al.* (2010), in short, the above discussion clearly identified the importance of LULCC in the climate system. It has also been established that biases, inaccuracies, and imprecision have been introduced to the climate monitoring systems because of meteorological station moves, instrument changes, improper exposure of instruments, and changes in observation practices (Davey and Pielke, 2005; Mahmood *et al.*, 2006). Hence, we also need strategies that will help us to detect and overcome these biases and thus lead to an improved understanding of the role of land-use forcing within the climate system (Mahmood *et al.*, 2010).

2.3.2 Global Warming

Land-use and land-cover changes are human activities that plays a dominant role, interact with the environment and have significant effects on the ecosystems at the local, regional, and global scales and consequently directly or indirectly exert great influence on the global climate changes (Foley *et al.*, 2005; Pachauri and Reisinger, 2007). People have come to realize that the global change always consists of a series of regional

changes with various processes and patterns according to the more and more observations (Fu *et al.*, 2003).

The continuous rise in temperature of the planet is really upsetting. The root cause for this is global warming. Global warming begins when sunlight reaches the earth. The clouds, atmospheric particles, reflective ground surfaces and surface of oceans then sends back about 30% of sunlight back into the space, whilst the remaining is absorbed by oceans, air and land. This consequently heats up the surface of the planet and atmosphere, making life feasible. As the earth warms up, this solar energy is radiated by thermal radiation and infrared rays, propagating directly out to space thereby cooling the earth. However, some of the outgoing radiation is re-absorbed by carbon dioxide, water vapours, ozone, methane and other gases in the atmosphere and is radiated back to the surface of the earth. These gases are commonly known as greenhouse gases due to their heat-trapping capacity.

On the other hand, since there is a great difference in the physical, chemical, and biological characteristics of the land surface in different regions on the earth, the impacts of the land-use and land-cover changes on the regional climate systems also vary greatly. For example, deforestation may lead to the temperature increase in the tropic zone, while it may lead to the regional temperature decrease in the Frigid Zone (Bonan, 2008). It has long been known that the land-use changes caused by human activities such as deforestation and agriculture practice have some effects on the climate (Kueppers and Snyder, 2012).

Owing to the limitation of knowledge and lack of interdisciplinary cooperation, there is still lack of full understanding of the mechanism of the procedures of the climate process and how the land-use changes influence the regional climate and consequently influence the global climate (Feddema *et al.*, 2005). There are separate studies of the biogeochemical and biogeophysical effects of the land-use changes on the climate in their respective fields, and there are few researches on their joint effects on the climate. However, there are still a lot of uncertainties in the current knowledge, and it is still very difficult to objectively assess the contribution of the land-use changes to the climate change (Arora and Montenegro, 2011). Hansen *et al.* (2005) proposed that the radiative forcing due to the change in albedo caused by the global land-use changes may decrease the temperature. By contrast, Intergovernmental Panel on Climate Change (IPCC) believed that the increase of CO₂ in the atmosphere will reinforce the radiative forcing, 1/3 of which was contributed by the land-use changes, and consequently increase the temperature (Pachauri and Reisinger, 2007). So, it is of great importance to objectively analyze the biogeochemical and bio-geophysical effects of the land-use changes on the climate (Foley *et al.*, 2005).

The planet has experienced the largest increase in surface temperature over the last 100 years. Between 1906 and 2006, the Earth's average surface temperature augmented between 0.6°C to 0.9°C, however out per year. Millions of pounds of methane gas are generated in landfills and agricultural decomposition of biomass and animal manure. Nitrous oxide is released into the atmosphere by various nitrogen-based

fertilizers including urea and diammonium phosphate and other soil management utilizations. Once released, these greenhouse gases stay in the atmosphere for decades or even longer. According to Intergovernmental Panel on Climate Change (IPCC), carbon dioxide and methane levels have increased by 35% and 148% since the industrial revolution of 1750.

2.3.3 Hydrological Cycle

Human impacts on global terrestrial hydrology have been accelerating during the 20th century. These human impacts include the effects of reservoir building and human water use, as well as land-cover change. To date, many global studies have focused on human water use, but only a few focuses on or include the impact of land-cover change. Results has shown that land-cover change has a strong effect on the global hydrological cycle in the same order of magnitude as the effect of human water use (applying irrigation, abstracting water, for industrial use for example, including reservoirs, etc.). When globally averaged, changing the land cover from that of 1850 to that of 2000 increases discharge through reduced evapotranspiration. The effect of land-cover change shows large spatial variability in magnitude and sign of change depending on, for example, the specific land-cover change and climate zone (Bosmans *et al.*, 2017).

Both emission-driven climate change as well as more direct human interventions such as dam building and water withdrawals (for domestic, industrial and agricultural use, including irrigation) have a strong impact on future water availability, floods and

droughts (e.g., Hirabayashi *et al.*, 2013; Haddeland *et al.*, 2014; Wanders and Wada, 2015; Winsemius *et al.*, 2016).

Overall, land-cover effects on evapotranspiration are largest for the transition of tall natural vegetation to crops in energy-limited equatorial and warm-temperate regions. In contrast, the inclusion of irrigation, water abstraction and reservoirs reduce global discharge through enhanced evaporation over irrigated areas and reservoirs as well as through water consumption. Hence, in some areas, land-cover change and water distribution both reduce discharge, while in other areas the effects may partly cancel out. The relative importance of both types of impacts varies spatially across climatic zones. (Bosmans *et al.*, 2017).

Additionally, humans have altered a large part of the land surface, replacing 41% (Sterling *et al.*, 2013), of natural vegetation by anthropogenic land cover such as crop fields or pasture. Such land-cover change can affect terrestrial hydrology by changing the evaporation-to-runoff ratio. To date, few studies focus on land-cover change when assessing the anthropogenic impact on the global terrestrial hydrological cycle (Bosmans *et al.*, 2017).

2.3.4 Forest Resources Degradation

Forest resources are such materials obtainable from forest lands to satisfy a need in production or direct utilization. Forest resources are grouped into three categories of

timber based, non-timber-based resources of plant origin and non-timber-based resources of animal origin (Lund, 2006).

Degradation results from over usage of resources to a level of low replenishment. Degradation is the reduction in supply and value of a naturally occurring resource. It refers to unfavourable alteration of the resource largely as a result of anthropogenic activities directly and/or indirectly (Osinem, 2005). There are currently more than 50 definitions of the concepts of degraded forest and forest degradation (Lund, 2009), with emphasis ranging from soil degradation (Hudson and Alcántara-Ayala, 2006), to more recently, loss of carbon stock and mitigation of climate change (Putz and Nasi, 2009; Morales-Barquero *et al.*, 2014; Stanturf *et al.*, 2014). A central aim of international negotiations and discussions surrounding forest degradation has therefore been to reach a consensus on these definitions (Putz and Redford, 2010; Armenteras and González, 2016; Chazdon *et al.*, 2016). International audiences continue to use the generic definition of “forest degradation” from FAO: “the reduction in the capacity of a forest to provide goods and services”. This definition is used to compare and monitor statistics across multiple countries, but does not include aspects such as a reference state or different degrees of degradation. As a consequence, Ghazoul *et al.* (2015) stated that interpreting degradation as a loss of attributes or functions, although intuitive, is not sufficient to establish the degree of degradation of a forest.

Thus, forest degradation refers to situations involving long-term and severe environmental changes, and does not cover short-term changes or variability such as

those associated with forest management for silvicultural purposes. The primary causes of forest degradation are unsustainable exploitation (in the form of excessive harvesting of forest products), overgrazing, wildfires, and the spread of invasive species or pests. Hosonuma *et al.* (2012) analyze the direct drivers of degradation in developing countries, and show that the main causes are harvesting for timber (52%), harvesting for firewood and charcoal (31%), uncontrolled forest fires (9%), and grazing (7%). Considering this variety of drivers, the process of degradation can be rapid, or may occur more slowly, such as the selective harvesting of the best trees in a forest, which may be conducted in few or through several occasions.

It is estimated that more than 2000 million hectares of the world's forest have been degraded (Stanturf *et al.*, 2014; Stanturf *et al.*, 2014). As a result, forest degradation is recognized globally as a serious environmental, social and economic problem (FAO, 2011; Simula and Mansur, 2011). The effects of forest degradation are harmful both to the ecosystems themselves and to society, as they have the potential to negatively affect millions of people who depend wholly or partially on the goods and services that forests generate on a local, regional or global scale (FAO, 2011).

Overall, it continues to be extremely difficult to establish a clear and practical definition of a degraded forest. This is in part due to the inherent level of uncertainty surrounding the degradation process but also stems from the high degree of variability in the capacity of forests to recover to their original, undisturbed state (i.e., resilience). Sasaki and Putz (2009) claim that given the need to reach agreements to address the

challenges posed by climate change, the definition of forest degradation should take into account the full range of biophysical and social conditions under which forests develop, and the variety of ways in which they can degrade.

2.3.5 Biodiversity Loss

Land uses can be regarded as intensive or extensive, depending on the degree to which they modify the environment. More intensive land uses result in a greater share of resources and energy flowing to human uses, leaving less to sustain other species. Hence, different land uses have varying impacts on the biodiversity of an area. The impact on biodiversity increases with the intensity of land use, from no impact to more than 95% of the mean species abundance (Taylor *et al.*, 2014). The loss of wildlife species from landscapes tends to occur, when clearing exceeds 20% of the landscape, and rapidly accelerates when less than 30% of the native vegetation remains (Morgan, 2001; McAlpine *et al.*, 2002). Cautioning against a simple threshold for all landscapes has shown that actual thresholds for decline may be affected by landscape productivity and the natural cover of the vegetation (Maron *et al.*, 2012). The impact of vegetation clearance on biodiversity may take decades to become apparent (McAlpine *et al.*, 2002).

2.4 Land-use/Land-over Change Detection

Digital land-use and land-cover change detection is the process of determining and/or describing changes in land-cover and land-use properties based on co-registered

multi-temporal remote sensing data. The basic premise in using remote sensing data for change detection is that the process can identify change between two (or more) dates that is uncharacteristic of normal variation. To be effective, change detection approaches must maximize inter-date variance in both spectral and spatial domains (i.e. using vegetation indices and texture variables). Numerous researchers have addressed the problem of accurately monitoring land-cover and land-use change in a wide variety of environments with a high degree of success (Chan *et al.*, 2001).

Change detection has emerged as a significant process in managing and monitoring natural resources and urban development mainly due to provision of quantitative analysis of the spatial distribution of the population of interest. Though, simple change detection is seldom adequate in itself, there is a requirement of information regarding initial and final land-cover types and uses, the “from-to” analysis. It is necessary to have accurate and up-to-date land-cover change information for understanding and assessment of the environmental consequences of such changes (Giri *et al.*, 2005).

Satellite Remote Sensing and GIS are the most common methods for quantification, mapping and detection of patterns of LULCC because of their accurate geo-referencing procedures, digital format suitable for computer processing and repetitive data acquisition (Lu *et al.*, 2004; Chen *et al.*, 2005). The basic purpose of using this data for detecting change is its ability to identify uncharacteristic change between two or more dates. Several researchers have addressed the problem of accurate monitoring of LULCC

in numerous environments (Chan *et al.*, 2001). The land-cover and land-use changes in agriculturally productive, arid and semi-arid land have been discussed in many studies. Rembold *et al.* (2000) used Landsat TM dataset for years 1972 and 1994 to calculate land-cover changes in lake regions of central/south Ethiopia. Shalaby and Tateishi (2007) identified types of LULC in the coastal zone of Egypt by making use of Landsat data. Gao and Liu (2010) performed digital analysis of two Landsat images to detect land-degradation trend that occurred at an interval of 10 years due to soil salinization and waterlogging in northeast China.

Numerous techniques have been developed and used for change detection to monitor changes in LULC by making use of remotely sensed data for instance image differencing, post-classification comparison (PCC), vegetation index differencing and principle components analysis (PCA) (Lu *et al.*, 2004). Among these, post-classification comparison was found to be the most accurate procedure by variety of studies as it offered an advantage of representation of nature of occurring changes. Thus, making use of PCC method minimizes associated problems with multi-temporal images that are recorded under different atmospheric and environmental conditions. It separately classifies the data from different dates, and thus, this multi-date data does not require any adjustment for direct comparison (Rivera, 2005; Zhou *et al.*, 2008; Warner and Campagna, 2009). An additional advantage of PCC method is the indication of the nature of change (Yuan *et al.*, 2005).

2.5 Normalized Difference Vegetation Index (NDVI)

The last half century has seen the development and use of different remotely sensed vegetation indices. The basic assumption behind the development and use of these indices is that some algebraic combination of remotely-sensed spectral bands can reveal valuable information such as vegetation structure, state of vegetation cover, photosynthetic capacity, leaf density and distribution, water content in leaves, mineral deficiencies and evidence of parasitic shocks or attacks (Jensen, 2007). The algebraic combination of spectral bands should therefore be sensitive to one or more of these factors. Conversely, a good vegetation index should be less sensitive to factors that affect spectral reflectance such as soil properties, atmospheric conditions, solar illumination, and sensor viewing geometry (Jensen, 2007).

2.6 Satellite Image Analysis

Satellite image analysis entails digital image processing, which involves manipulation and interpretation of the digital image data by specific computer programs to display and extract meaningful information about the surface of the earth. Digital-image classification which is among the basic image analysis processes governs most of the LULC change detection study (Matinfar *et al.*, 2007). The satellite sensors record the variation in the electromagnetic radiation from each part of the earth surface and assign it with a distinct DN value for each spectral band (Oumer, 2009). The range of digital number varies from sensor to sensor and depends on the radiometric resolution, which is

attributed to the sensor's sensitivity to various level of incoming energy (Ayele, 2011). The variation in the spectral reflectance of a particular LULC type is captured during the digital image classification process and thus the LULC map of an area is constructed. For instance, the spectral signature of water is different from that of vegetation for each band of a multi-spectral imagery and vice-versa.

2.7 Land-use Modeling

Sohl *et al.* (2010) discussed the need to address several “foundational elements” of LULC modeling, which includes, geographic context, regional land-use history, representation of drivers of change and representation of local land-use patterns. Heistermann *et al.* (2006) noted four classes of data needed for LULC models, three of which can be derived from or supported by remote-sensing data, they include, current and historical land-use data, environmental data and scenario data. Several of these foundational elements can be directly informed through the use and analysis of remotely sensed data.

Remote Sensing and GIS along with computer-based modeling tools have become a popular and efficient means of simulating current and future land-use and land-cover change and hence, assist in land-use planning and natural-resources management (Herold *et al.*, 2003; Araya, 2009). Various tools exist in the era of land-use modeling which are diverse in design, spatial scale, temporal dimension, data availability etc. The Markov Chain model is a unique and widely used tool in land-use modeling which demonstrates

the LULC changes as a stochastic process (Weng, 2002). In the Markovian system, the future state of a land-use system is modeled on the basis of the immediate proceeding state (Araya, 2009). The Markov Chain analysis describes the probability of LULC changes from one period to another by constructing a transition probability matrix between period-1 and period-2.

2.8 Application of Remote Sensing and GIS

2.8.1 Remote Sensing

Since the launch of Landsat-1, the first Earth resource satellite in 1972, remote sensing has become an increasingly important tool for the inventory, monitoring, and management of earth resources. The increasing availability of information products generated from satellite imagery data has added greatly to our ability to understand the patterns and dynamics of the earth resource systems at all scales of inquiry (Bawahidi, 2005).

Viewing the earth from space is now crucial to the understanding of the influence of man's activities on natural resources base over time. Hence remote sensing data and technique and geographic information systems (GIS) provide efficient methods and opportunities for analysis of land-use and land-cover dynamics and tools for land-use planning and modeling (Miller *et al.*, 2007). A particularly important application of remote sensing is the generation of land-use/land-cover maps from satellite imagery. Compared to more traditional mapping approaches such as terrestrial survey and basic

aerial photo-interpretation, land-use mapping using satellite imagery has the advantages of low cost, large area coverage, repetitively, and computability (Franklin, 2001).

Consequently, land-use information products obtained from satellite imagery such as land-use maps, data and GIS layers have become an essential tool in many operational programs involving land-resources management. The prospect for the use of satellite-imagery data in land-use management and planning is an extremely promising one. As a result of the recent development of sensor technology, the quality of satellite imagery available for land-use mapping is improving rapidly. Particularly noteworthy in this regard is the improved spatial and spectral resolution of the imagery captured by new satellite sensors (Bawahidi, 2005).

The use of imagery from high-resolution sensors on satellites such as IKONOS and QuickBird has proved that data from space-borne sensors can provide a viable alternative to aerial photography in many applications including detailed land-cover mapping, water-resources assessment, irrigation management and, crop and yield mapping (Treitz and Rogan, 2004).

Remote-sensing data and analyses are a vital component of many LULC-modeling efforts. Given the reliance on remote-sensing data for informing LULC models, a primary concern for modelers is continued availability of consistent data (Sohl and Sleeter, 2012). Heistermann *et al.* (2006) noted that a major problem in LULC modeling was the availability of spatially explicit time series data, while Rindfuss *et al.* (2004) stated that it was often impossible to obtain consistent, cost-effective, spatially and

temporally relevant historical remote-sensing data for supporting land-change science. Crews and Walsh (2009) noted the extreme importance of continued development and maintenance of consistent satellite sensors and databases, with timely data delivery at reasonable prices.

2.8.2 Geographical Information Systems (GIS)

Goodchild (2000) defines GIS as a computing application that allows the user to create, store, manipulate, visualize and analyze geographic information. The strongest fields in which GIS proves to be most useful is resources management, utilities management, telecommunications, urban and regional planning, vehicle routing and parcel delivery as well as all the sciences which involves the surface of the earth.

A GIS provides a framework for gathering and organizing spatial data and related information which can be displayed and analyzed. Therefore, GISs does not only have a central database, but it can also be used for organizing, exploring, analyzing, editing and viewing geographical data. Geographical data can range from a set of single geo-referenced features or point observations to more complex collections of data in database structures (Goodchild *et al.*, 2007).

In the past years, new technology and satellite data provided geodesists with a better understanding and measurements to calculate the best earth-fitting spheroid. Coordinates are related to the earth's center of mass, generating a geocentric datum. Satellite technology enabled cartographers to measure the earth's shape more accurately.

Therefore, a new ellipsoid was calculated called the World Geodetic Survey (WGS 1984). The reference point used today in the South African implementation is the Hartbeeshoek Radio Astronomy Observatory near Pretoria. The Hartbeeshoek 1994 datum was officially implemented on the 1st of January 1999. This is basically a local implementation of WGS 84 (Thomas, 2007). The WGS 84 datum is a combination of reference points and the WGS 1984 ellipsoid designed to fit areas around the world. In South Africa, the differences between the Hartbeeshoek 94 and WGS 84 datum is very small. The most common used datum is WGS 1984 which serves as a framework for locational measurements across the globe (Kennedy and Kopp, 2000).

Projected coordinate systems are developed for flat, two-dimensional surfaces. Projected coordinate systems have even lengths, angles and areas on a flat two-dimensional surface. Locations on the earth's surface are identified by x,y coordinates on a grid, starting at the center of the grid. Each object or specific location has two values that determine the central position named the horizontal and vertical values (also called the x-coordinate and y-coordinate). When an object has to be located the user starts at the center of the grid where the x and y values are zero and then move towards the location on the x and y axis of the grid. A projected coordinate system is always based on a geographic coordinate system derived from a sphere or spheroid (Kennedy and Kopp, 2000).

Rich *et al.* (2010) conducted a Building Interior Space Data Model (BISDM) for GIS in the summer of 2007. They described GIS as a core technology used to build the

information systems such as cartographic, image, business analysis and faculty management. GIS is used to manage, analyze and report building data at all scales. The scales range from global, country, region, city, campus, building to the rooms inside the buildings.

2.9 Sustainable Forest Management

While avoiding deforestation and forest degradation may help meet short term goals, sustainable forest management aimed at providing timber, fiber, biomass and non-timber resources can provide long-term livelihood for communities, can reduce the risk of forest conversion to non-forest uses (settlement, crops, etc.), and can maintain land productivity, thus reducing the risks of land degradation (Gideon *et al.*, 2014; Nitcheu *et al.*, 2016; Sufo *et al.*, 2016).

Developing sustainable forest management strategies aimed at contributing towards negative emissions throughout this century requires an understanding of forest management impacts on ecosystem carbon stocks (including soils), carbon sinks, carbon fluxes in harvested wood, carbon storage in harvested wood products and the emission reductions achieved through the use of wood products (Lemprière *et al.*, 2013; Kurz *et al.*, 2016). Transitions from natural to managed forest landscapes can involve a reduction in forest carbon stocks, the magnitude of which depends on the harvest rotation length relative to the frequency and intensity of natural disturbances (Harmon *et al.*, 2009), but can increase the forest carbon sink, and generate climate benefits through increasing the

stock of wood products and displacing GHG-intensive building materials (Werner *et al.*, 2010; Smyth *et al.*, 2014).

In boreal forest ecosystem, measurements of carbon stocks and carbon fluxes indicate that old growth stands are typically small carbon sinks or carbon sources (Hadden and Grelle, 2016; Gao *et al.*, 2018). Age-dependent increases in forest carbon stocks and declines in forest carbon sinks mean that landscapes with older forests have accumulated more carbon but their sink strength is diminishing, while landscapes with younger forests contain less carbon but they are removing CO₂ from the atmosphere at a much higher rate (Volkova *et al.*, 2017). This fundamental trade-off between maximizing forest carbon stocks and maximizing ecosystem carbon sinks is at the origin of ongoing debates about optimum management strategies to achieve negative emissions (Keith *et al.*, 2014; Lundmark *et al.*, 2014; Kurz *et al.*, 2016). With increasing forest age, carbon sinks in forests will diminish until harvest or natural disturbances such as wildfire remove biomass carbon or release it to the atmosphere.

Sustainable forest management, including harvest and forest regeneration, can help maintain active carbon sinks by maintaining a forest age-class distribution that includes a share of young, actively growing stands (Volkova *et al.*, 2018). The use of the harvested carbon in either long-lived wood product (e.g. for construction), short-lived wood products (e.g., pulp and paper), or biofuels affects the net carbon balance of the forest sector. The use of these wood products can further contribute to GHG emission

reduction goals by avoiding the emissions from the products that have been displaced (Lemprière *et al.*, 2013).

Initial landscape conditions, in particular the age-class distribution and therefore carbon stocks of the landscape strongly affect the mitigation potential of forest management options (Kilpeläinen *et al.*, 2017). Landscapes with predominantly mature forests may experience larger reductions in carbon stocks during the transition to managed landscapes (Harmon *et al.*, 2009), while in landscapes with predominantly young or recently disturbed forests, sustainable forest management can enhance carbon stocks. Sustainable forest management, including the intensification of carbon-focused management strategies, can make long-term contributions towards negative emissions if the sustainability of management is assured through appropriate governance, monitoring and enforcement.

CHAPTER THREE

3.0

METHODOLOGY

3.1 Study Area

The study was carried out in Aworo Forest Reserve, located in Egbado North Local Government Area of Ogun State, Nigeria. The forest reserve covers an area of about 24,906 ha of land, and it lies within latitudes $7^{\circ}10'30''$ and $7^{\circ}21'00''$ N and longitudes $2^{\circ}43'00''$ and $2^{\circ}58'00''$ E (Figure 1).

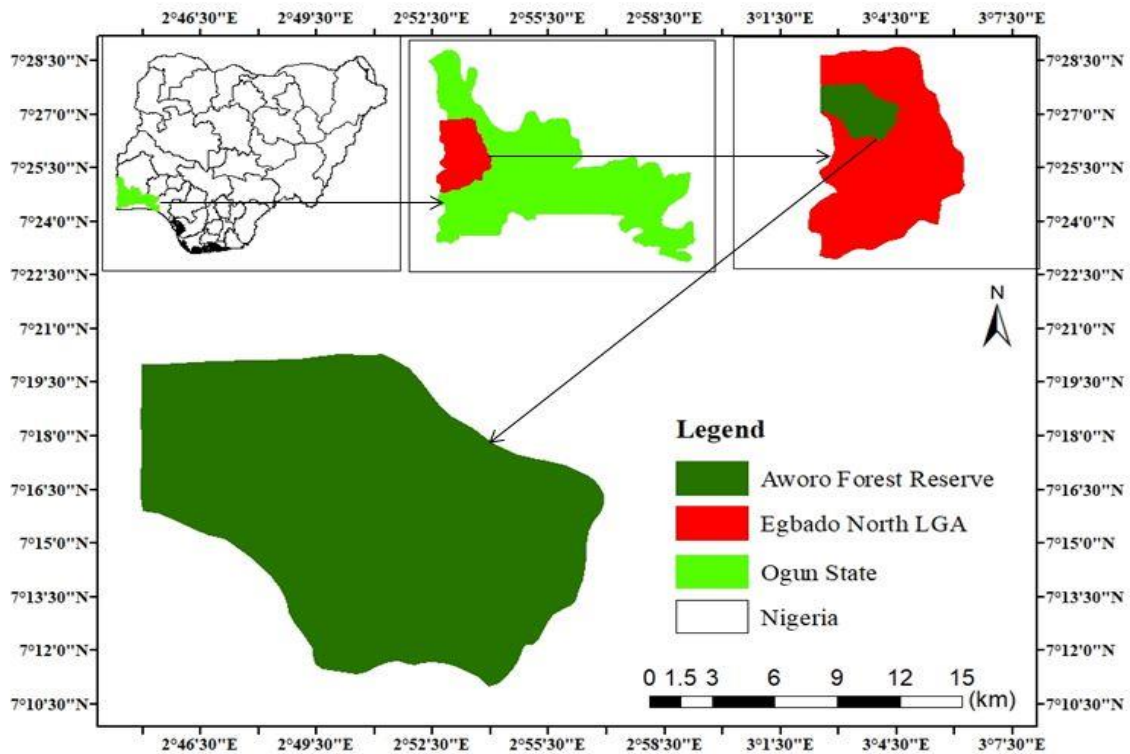


Figure 1: Map of the study area

3.1.2 Climate

The climate of the study area is sub-humid tropical with two main climatic conditions. The annual rainfall normally spreads over eight months, between April and November, and ranges between 100 and 200 mm, having bi-modal pattern with peaks in May/June and September/October (Oluwalana, 1997). The relative humidity is high all year round and generally above 80% during the wet season and ranges between 60% and 80% during the dry season. The average daily maximum temperature varies from 28°C in the wet season and 32°C in the dry season.

3.1.3 Geology, Climate and Vegetation

Aworo Forest Reserve is a semi-savannah vegetation area endowed with favourable climatic conditions for agricultural pursuit throughout the year. Its tropical nature is also an added advantage for agriculture. The forest reserve has an elevation of 160 m and a land area of about 24,906 ha.

3.2 Sampling Procedures and Data Collection

3.2.1 Selection and Acquisition of Images

The study is based on the analysis of satellite imagery of the study area. Therefore, the first task was the selection of satellite sensors and associated images. During this process, prior considerations were the purpose of the study, objects to be identified, and the availability of images.

After considering all of these factors, five Landsat images were selected for analysis, and acquired from the USGS Earth Resources Observation and Science Center (EROS). All acquired images were dated from late December, to early February, which corresponds to dry season, to keep the analysis free from seasonal variability impact and excessive cloud cover. Individual scene was obtained from path 191 and row 55. Table 1 summarizes the image data descriptions.

Table 1: Description of satellite imageries used in LULC change detection

Year	Date	Satellite Imagery	Spectral resolution	Spatial resolution
1984	December, 18	Landsat 5 (TM)	Band 1 to 7	30m
2000	February, 06	Landsat 7 (ETM)	Band 1 to 7	30m
2013	December, 18	Landsat 8 (OLI/TC)	Band 1 to 7	30m
2020	January, 20	Landsat 8 (OLI/TC)	Band 1 to 7	30m
2023	January, 25	Landsat 8 (OLI/TC)	Band 1 to 7	30m

Landsat data are multi-spectral with seven different colour bands although the Landsat Enhanced Thematic Mapper Plus (ETM+) sensor has an additional panchromatic band 8, which was not used in this study. For both TM and ETM+ sensors, Table 2 describes, in detail, the spectral features of different Landsat bands.

Table 2: Spectral features of Landsat image bands

Band Number	Wavelength Interval (µm)	Spectral Response
1	0.45-0.52	Blue-Green
2	0.52-0.60	Green
3	0.63-0.69	Red
4	0.76-0.90	Near IR
5	1.55-1.75	Mid-IR
6	10.40-12.50	Thermal IR
7	2.08-2.35	Mid-IR

3.3 Ground Truthing

Validating accuracy of land-use classes interpreted from remote sensing imageries requires a validated superior resolution dataset of the same area. Nevertheless, in a place such as the study area, where superior data is not available, fieldwork remains the best option. The field check was conducted using a hand-held Global Positioning System (GPS) receiver.

According to Jensen (1986), the ideal number of checkpoints required to be tested in the land-use classification map is determined from the binomial probability, using:

$$N = \frac{4 (p)(q)}{e^2} \dots \dots \dots (1)$$

Where:

p = is the expected percentage accuracy;

$q\sim$ = the difference between 100 and p ;

e = is the maximum allowable error.

For an expected 90% accuracy and 5% allowable error, the minimum number of points required is 144. The checkpoints were downloaded from the GPS device and imported into ArcGIS, and added to the GIS database, as an event theme. This was converted into data layer. This theme of field coordinates was then used as a basis for assessing accuracy of the classified imageries.

3.4 Image Analysis

3.4.1 Image Processing

The acquired images were processed and analyzed in ArcGIS 10.8 using the Geo-tiff format for further processing and enhancement. The first operation was the compositing of the images. Bands of interest were selected and stacked. From the stacked bands, a colour composite of bands 4, 3, 2, and 5, 4 and 3 were generated and resampled for Landsat TM/ETM+ and OLI, respectively. These combinations have been regarded as efficient and adequate, when using Landsat data for studying land-use and land-cover change, especially, if it has to do with vegetation, farmland, water bodies, wetland, bare surface and built-up area.

Figure 2 shows the composite images of Aworo Forest Reserve between 1984 and 2023. Bands 4, 3 and 2 (Red, Green and Blue filters, respectively) were selected and composited from the imagery of 1984 (Landsat 5, TM) and 2000 (Landsat 7, ETM). Similarly, bands 5, 4 and 3 were selected and composited from the imagery of 2013 (Landsat 8, OLI), 2020 (Landsat 8, OLI) and 2023 (Landsat 8, OLI). These filters are a reversal of the true colour band combination, resulting in a false colour band combination with a basic aim of creating a visual map of the area. It helps in visualization of wavelengths that the human eye cannot see (i.e. near-infrared and beyond).

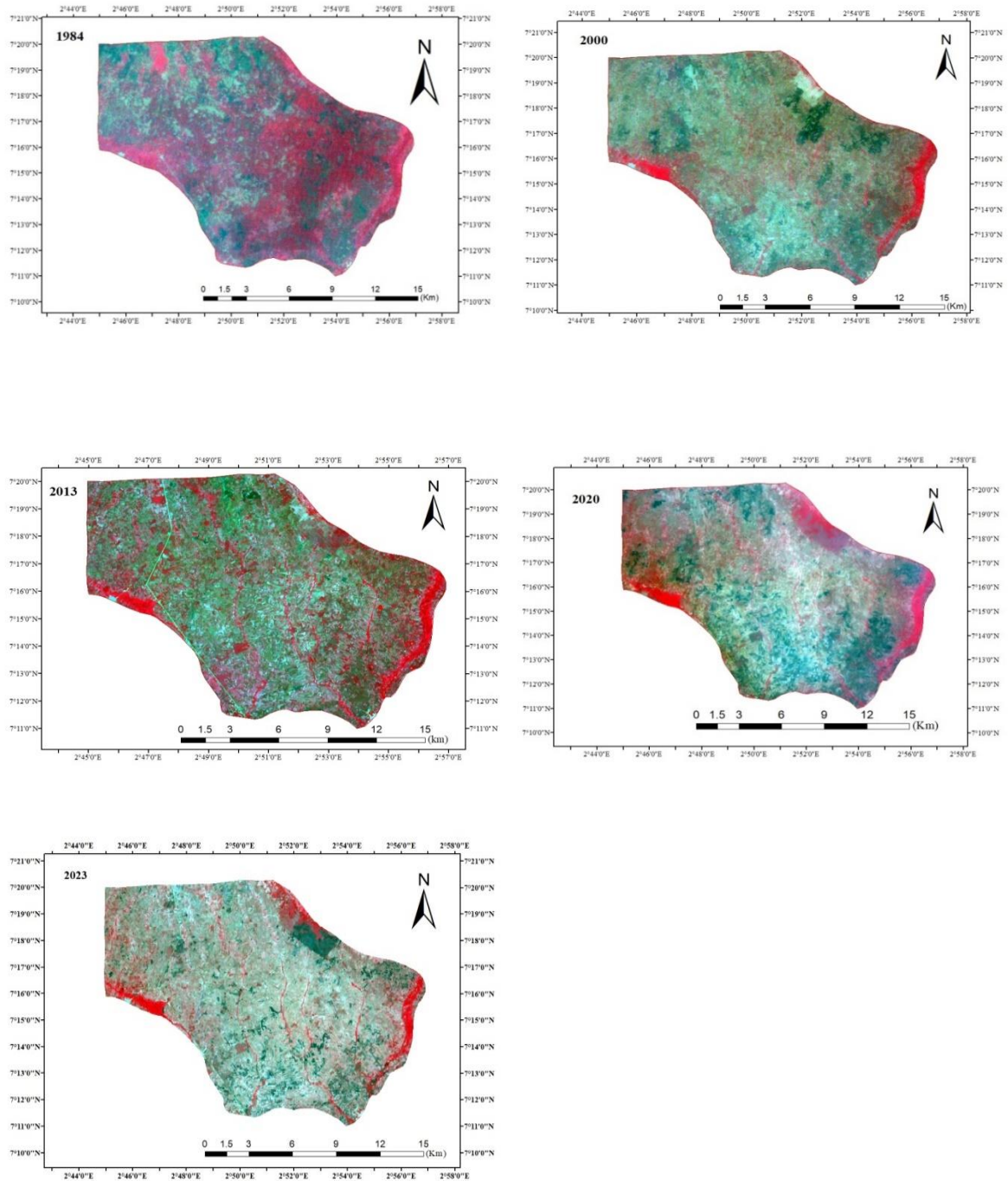


Figure 2: Composite images of the study area for the five datasets

3.4.2 Image Classification

Using ArcGIS 10.8, the area of interest was extracted from the image. To commence the image classification, a composite image was created for each of the periods. The Iso-cluster classification technique was used to determine the land-use/land-cover classes, after which a reclassification was carried out to deduce the exact land-use and its corresponding area coverage.

3.4.3 Land-use/Land-cover Change Analysis

Change detection for land use and land cover was carried out using pre-classification and post-classification approaches. The pre-classification change detection involved matching pixels for pixels to process multi-date imageries of the same area so as to generate changes.

As a way of determining the land-use/land-cover change (LULCC) and future predictions, land-use/land-cover layers of the selected dates were overlaid with one another in the ArcGIS environment. Change analysis was then performed by intersecting the different multi-temporal land-use/land-cover layers (for the selected dates). A contingency matrix of change was then developed using a pivot table. This was done to examine the land-use and land-cover changes of Aworo Forest Reserve within the study period.

The record (row) totals indicate the area extent of each LULC class in the base year (1984) and the field (column) totals represent the area of each LULC class in current

year (2023). This means that, by reading down each row of the table, the transition (loss) of the row header class into other LULC classes in year 2023 are indicated. Also, by reading down each column, the transition (gain) from each LULC class to the column header class in year 2023 is indicated. Therefore, the column represents gains from other LULC classes to the column header, while the row represents loss of row headers to other LULC classes.

3.5 Vegetation Indices

The Normalized Difference Vegetation Indices (NDVI) is the most widely and frequently used vegetation indices. The combination of its normalized difference formulation and the use of highest absorption and reflectance region of chlorophyll make it robust over a wide range of conditions. The NDVI is defined by the following equation;

$$NDVI = \frac{\rho_{NIR} - \rho_{RED}}{\rho_{NIR} + \rho_{RED}} \dots \dots \dots (2)$$

The values of NDVI range between -1 and 1. The common range for green vegetation is 0.2 to 0.8. The NDVI was computed for the classified Landsat image of all study dates, using the corresponding bands

3.6 Trend Analysis

The magnitude of change (area) and the percentage change for each study period were evaluated based on Adeyemi and Ayinde (2022), as follows:

$$\text{Area} = \text{Count} \times 0.0009 \dots \dots \dots (3)$$

$$\text{Percentage} = \frac{\text{Count}}{\text{Sum of Count}} \times 100 \dots \dots \dots (4)$$

The change in magnitude, percentage trend and annual rate of change for each LULC class was calculated as:

$$\text{MC} = A_f - A_i \dots \dots \dots (5)$$

$$T = \left(\frac{A_f - A_i}{A_i} \times 100 \right) \dots \dots \dots (6)$$

$$R = \left(A_f - A_i \times \frac{1}{t} \right) \dots \dots \dots (7)$$

$$\text{ARC} = \left(\frac{A_f - A_i}{A_i} \times \frac{1}{t} \right) \times 100 \dots \dots \dots (8)$$

Where:

A_f = area in final year;

A_i = area in base/initial year;

MC = magnitude of change (ha);

T = trend (%);

R = rate of change (ha);

t = periodic interval;

ARC = annual rate of change (%)

3.7 Accuracy Assessment

Confusion matrix was computed to assess the image classification accuracy of the land-use/land-cover classes for 2023. The GCP data collected were used for image classification and overall accuracy assessment in combination with the classified raster image. The overall accuracy evaluated the relationship between the remote sensing-derived map and the assumed true map in total area in each land-use category. However, it does not provide for compensating errors that occur in the various categories. Specific errors of omission and commission on each land-use class are required to compliment the overall estimated accuracy. The GCPs were analyzed for error matrix using the omission and commission error computation method. omission and commission errors were calculated for the different land-use and land-cover classes, as well as producer's and user's accuracy. The accuracy for each class was estimated, and the accuracy statistics was also determined.

$$P = \frac{\sum_{i=1}^k n_{ii}}{n} \times 100 \dots \dots \dots (9)$$

Where:

- n_{ii} = number of correct points;
- n = total number of ground control points;
- P = overall accuracy.

$$\text{Sensitivity} = \frac{a}{a + b} \text{ (equivalent to Producer's Accuracy) } \dots \dots \dots (10)$$

$$\text{Specificity} = \frac{d}{b + d} \dots \dots \dots (11)$$

$$\text{Commision error} = 1 - \text{Specificity} \dots \dots \dots (12)$$

$$\text{Ommision error} = 1 - \text{Sensitivity} \dots \dots \dots (13)$$

$$\text{Positive Predictive Power} = \frac{a}{a + b} \text{ (Equivalent to User's accuracy).} \dots \dots \dots (14)$$

$$\text{Negative Predictive Power} = \frac{d}{c + d} \dots \dots \dots (15)$$

Where:

a = number of times a classification agreed with the observed value;

b = number of times a point was classified as X when it was observed to not be X;

c = number of times a point was not classified as X when it was observed to be X;

d = number of times a point was not classified as X when it was not observed to be X;

Total points (accurately classified) = $n = (a + b + c + d)$.

The Khat statistics yielded by Kappa analysis is computed as:

$$K_c = \frac{N \sum_{i=1}^r x_{ii} - \sum_{i=1}^r (x_{i+} \times x_{+i})}{N^2 - \sum_{i=1}^r (x_{i+} \times x_{+i})} \dots \dots \dots (16)$$

Where:

K_c = Overall Kappa's coefficient;

r = number of rows and columns in error matrix;

N = total number of observations (pixels);

X_{ii} = observation in row i and column i ;

X_{i+} = total for row i ;

X_{+i} = total for column i .

A Kappa coefficient equal to 1 means perfect agreement, whereas a value close to zero means that the agreement is no better than would be expected by chance. The categorization of Kappa statistic based on similar reproduction done by Rwanga and Ndambuki (2017) is presented in Table 3.

Table 3: Rating criteria of Kappa statistics

S/N	Kappa statistics	Strength of agreement
1	<0.00	Poor
2	0.00 – 0.20	Slight
3	0.21 – 0.40	Fair
4	0.41 – 0.60	Moderate
5	0.61 – 0.80	Substantial
6	0.81 – 1.00	Almost perfect

Source: Rwanga and Ndambuki (2017)

3.8 Future Projection

Markov chain analysis was used for simulating future scenarios of land cover for the study area. Markov chain analysis describes land-use change from one period to another and uses this as the basis to project future changes (Zubair, 2006). This was used to predict the status of each land-use/land-cover type in the future, based on the current situations/scenarios.

$$F = A \times x_i + C \dots \dots \dots (17)$$

Where:

F = future trend;

A= annual rate of change;

x_i = number of interval (years) for the ith year being predicted;

C = current year area magnitude

CHAPTER FOUR

4.0

RESULTS

4.1 Land-use Classes of Aworo Forest Reserve between 1984 and 2023

The classification of the time-series images yielded five land-use types for the study area. These land-use and land-cover types are described in Table 4.

Table 4: Description of land-use class of Aworo Forest Reserve

LULC	Description
Primary forest	This refers to areas of dense vegetation cover and closed canopy, where there is no clearly-visible indication of human activities.
Secondary forest	This is a forest or woodland area which has re-grown after a timber harvest or clearing for agriculture.
Agricultural land	It includes areas with farming activities including arable crops, agricultural tree cover and cash cropping e.g., cassava, maize, etc.
Fallow land	This refers to all areas previously cultivated and left temporarily uncultivated for the time being.
Bareland/built-up	These refer to areas with little or no account of vegetation-cover, and include settlements and other human infrastructural developments.

Table 5 presents the spatial extent of LULC in 1984, 2000, 2013, 2020 and 2023, while Figure 3 shows the LULC maps of Aworo Forest Reserve during the same period. The land-use classification for the year 1984 in the study area showed that fallow land had the highest area of 7,932 ha, approximately 31.8% of the reserve. Agricultural land had a total extent of 5,874 ha, which was about 23.6% of the reserve. This was followed by secondary forest with 5,446 ha (21.9%), then, primary forest with 4,290 ha (17.2%) while bareland/built-up had 1,364 ha (5.5%).

In 2000, agricultural land and bareland/built-up had increased in area to about 9,943 ha and 3,937 ha, representing 40.0% and 15.8%, respectively, while primary forest, secondary forest and fallow land experienced a decreased to about 676 ha (2.7%), 3,245 ha (13.0%) and 7,105 ha (28.5%), respectively.

In 2013, there was a decrease in agricultural land to about 9,836 ha, approximately (39.5%) of the area while primary forest and Secondary forest increased to 2,877 ha (11.6%) and 5,682 ha (22.8%), respectively. Bareland/built-up also decreased to 3,477 ha (13.9%), while the fallow land occupied 3,034 ha (12.2%) in that year.

By 2020, secondary forest and agricultural land had gained increase to about 6,872 ha and 11,257 ha, respectively, from the previous year, which corresponds to 27.6% and 45.2%, respectively. Bareland/built-up occupied 3,321 ha (13.3%). However, fallow land and primary forest had reduced to 1,734 ha (7.0%) and 1,722 ha (6.9%), respectively.

In 2023, the forest reserve was dominated by agricultural land, covering about 11,020 ha (44.2%). This was followed by secondary forest, occupying about to 5,887 ha (23.6%). An increase was observed in the case of primary forest and fallow land to about 2,404 ha (9.7%) and 3,430 ha (13.8%), respectively while bareland/built-up decreased to 2,165 ha (8.7%).

Table 5: Land-use and land-cover spatial extents of AFR between 1984 and 2023

	1984		2000		2013		2020		2023	
LULC	Area		Area		Area		Area		Area	
	(ha)	(%)	(ha)	(%)	(ha)	(%)	(ha)	(%)	(ha)	(%)
PF	4,290	17.2	676	2.7	2,877	11.6	1,722	6.9	2,404	9.7
SF	5,446	21.9	3,245	13.0	5,682	22.8	6,872	27.6	5,887	23.6
AL	5,874	23.6	9,943	40.0	9,836	39.5	11,257	45.2	11,020	44.2
FL	7,932	31.8	7,105	28.5	3,034	12.2	1,734	7.0	3,430	13.8
BU/BL	1,364	5.5	3,937	15.8	3,477	13.9	3,321	13.3	2,165	8.7
Total	24,906	100	24,906	100	24,906	100	24,906	100	24,906	100

N.B.: PF = primary forest; SF = secondary forest; AL = agricultural land; FL = fallow land; BL/BU = bareland/built-up

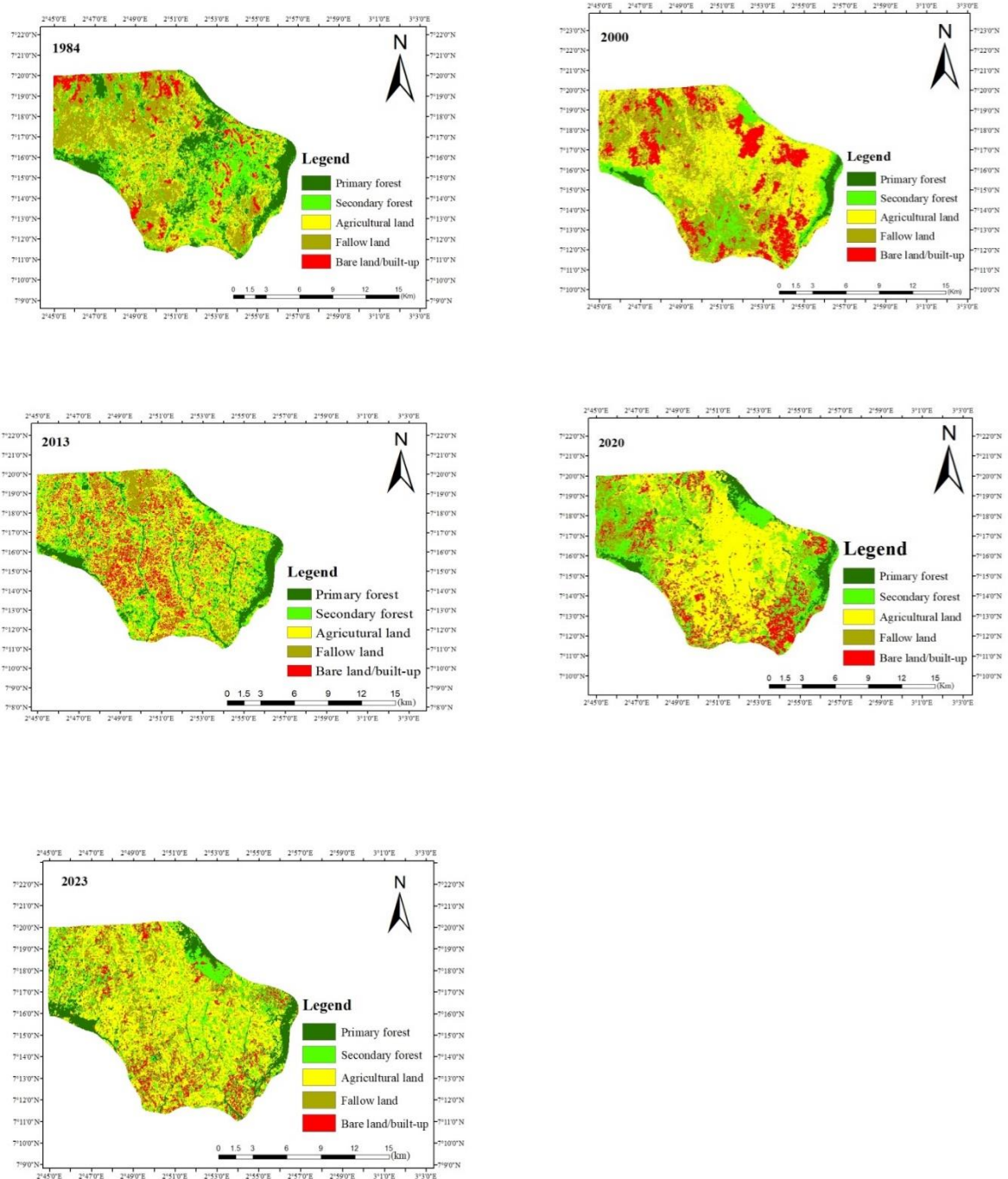


Figure 2: Land-use/land-cover maps of Aworo Forest Reserve between 1984 and 2023

Figure 4 presents the spatial extents of the LULC areas in the study years. Figure 5 presents the NDVI maps of Aworo Forest Reserve between 1984 and 2023. The highest value for the NDVI (0.46) was in 2013, which indicates the increase in vegetation and corresponds to the increase in primary forest in that year. The lowest NDVI value (-0.34) was in 2000, which corresponds to the significant loss in vegetative cover in that year (Table 6).

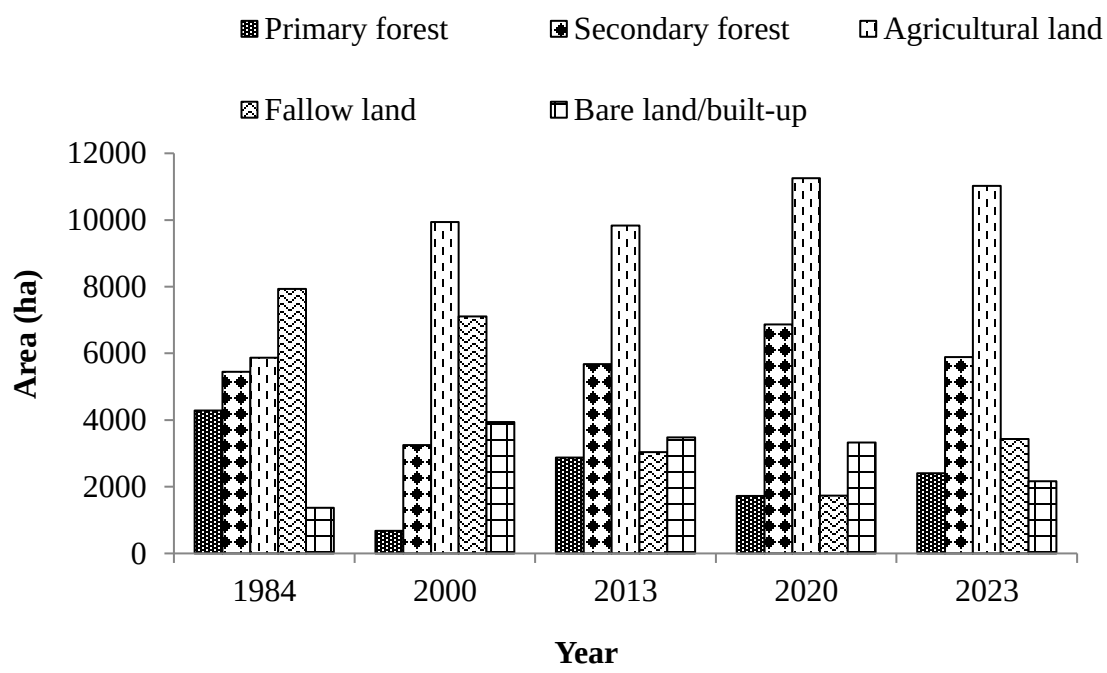


Figure 3: Spatial extents of LULC in AFR between 1984 and 2023

Table 6: Descriptive statistics for the NDVI analysis between 1984 and 2023

Year	NDVI statistics		
	Minimum	Maximum	Mean $\pm$ SD
1984	-0.02	0.28	0.12 $\pm$ 0.10
2000	-0.34	0.15	-0.16 $\pm$ 0.18
2013	0.05	0.46	0.25 $\pm$ 0.14
2020	0.04	0.33	0.17 $\pm$ 0.10
2023	0.04	0.36	0.18 $\pm$ 0.11

N.B.: SD = Standard Deviation

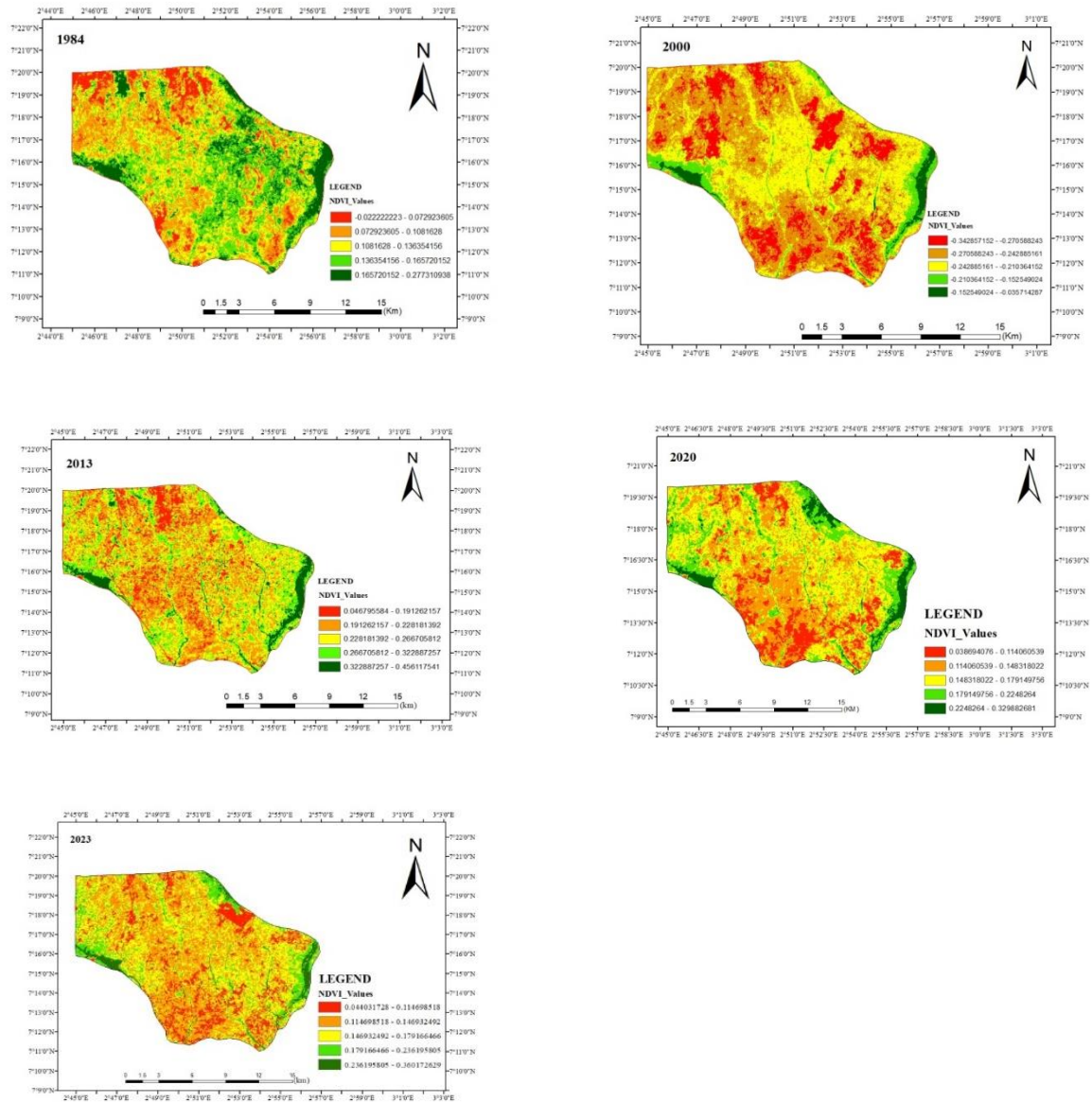


Figure 4: Maps of NDVI in AFR between 1984 and 2022

4.2 Land-use/cover Changes in Aworo Forest Reserve between 1984 and 2023

The changes in the land-use and land-cover types of Aworo Forest Reserve between 1984 and 2023 are presented in Table 7 and Figure 6. The signs (+) and (-) indicate increase and decrease, respectively. The magnitude of change in LULC between 1984 and 2000 showed an increasing area coverage for agricultural land (4,069 ha) and bareland/built-up (2,573 ha), while primary forest (-3,614 ha), secondary forest (-2,201 ha) and fallow land (-826 ha) experienced losses.

Between 2000 and 2013, primary forest and secondary forest increased to about 2,201 ha and 2,437 ha, respectively, while fallow land (-4,071 ha), bareland/built-up (-460 ha) and agricultural land (-107 ha) experienced a decrease.

However, between 2013 and 2020, there was an increase in agricultural land and secondary forest to 1,421 ha and 1,190 ha respectively, while a decline was recorded for primary forest (-1,156 ha) as well as fallow land (-1,300 ha) and bareland/built-up (-156 ha).

The changes within the study period of 2020 to 2023 showed increases in fallow land to (1,696 ha) and primary forest to (628 ha) while bareland/built-up (-1,157 ha), secondary forest (-985 ha) and agricultural land (-238 ha) experience losses within the period.

The changes over the entire study period (1984-2023) revealed losses in primary forest (-1,886 ha) and fallow land (-4,502 ha), in contrast, an increase was observed in

agricultural land (5,146 ha), bareland/built-up (801 ha) and secondary forest (441 ha). Details of changes are shown in Table 7.

Table 7: Magnitude of change of LULC in hectares (ha) between 1984 and 2023

Period	PF (%)	SF (%)	AL (%)	FL (%)	BL (%)
1984 - 2000	-3,614 (-84.2)	-2,201 (-40.4)	4,069 (69.3)	-826 (-10.4)	2,573 (188.7)
2000 - 2013	2,201 (325.6)	2,437 (75.1)	-107 (-1.1)	-4,071 (-57.3)	-460 (-11.7)
2013 - 2020	-1,156 (-40.2)	1,190 (21.0)	1,421 (14.4)	-1,300 (-42.8)	-156 (-4.5)
2020 - 2023	682 (39.6)	-985 (-14.3)	-238(-2.1)	1,696 (97.8)	-1,157 (-34.8)
1984 – 2023	-1,886 (-40.0)	441 (8.1)	5,146 (87.6)	-4,502 (-56.8)	801 (58.7)

N.B.: PF = primary forest; SF = secondary forest; AL = agricultural land; FL = fallow land; BL/BU = bareland/built-up

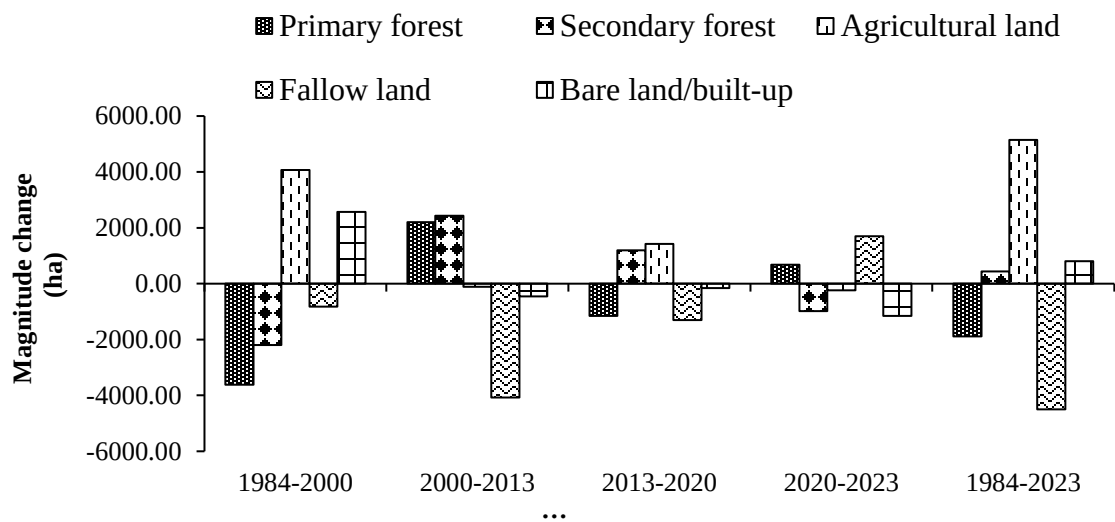


Figure 5: Magnitude of changes of LULC in AFR between 1984 and 2023

The annual rates of change for the LULC as observed in Aworo Forest Reserve within the study period are presented in Table 8. Between 1984 and 2000, a gain of 11.8% was observed for bareland/built-up and 4.3% for agricultural land, while the primary forest, secondary forest and fallow land lost -5.3%, -2.5% and - 0.7% per annum, respectively.

The changes observed between 2000 and 2013 were a loss of fallow land, bareland/built-up and agricultural land by -4.4%, -0.9% and -0.1%, respectively while secondary forest and primary forest increased by 5.8% and 25.0%, respectively.

The changes observed between 2013 and 2020 shows that there were losses in fallow land (-6.1%), primary forest (-5.7%) and bareland/built-up (-0.6%). However, increasing trends were recorded in agricultural land and secondary forest to 2.1% and 3.0%, respectively.

The changes between 2020 and 2023 showed annual losses in secondary forest, agricultural land and bareland/built-up to about -7.2%, -1.1% and -17.4%, respectively, while primary forest and fallow land increased to about 19.8% and 48.9%, respectively.

The trend over the entire study period (1984-2023) also revealed that there were annual losses for primary forest (-1.2%) and fallow land (-1.5%), while agricultural land (2.3%), bareland/built-up (1.5%) and secondary forest (0.2%) experienced increase. Details are presented in Figure 7.

Table 8: Annual rate of change (ha) for LULC in AFR between 1984 and 2023

Period	PF (%)	SF (%)	AL (%)	FL (%)	BL (%)
1984 – 2000	-225.9 (-5.3)	-137.6 (-2.5)	254.3 (4.3)	-51.7 (-0.7)	160.8 (11.8)
2000 – 2013	169.3 (25.0)	187.4 (5.8)	-8.2 (-0.1)	-313.2 (-4.4)	-35.4 (-0.9)
2013 – 2020	-165.1 (-5.7)	170.0 (3.0)	203.0 (2.1)	-185.7 (-6.1)	-22.3 (-0.6)
2020 – 2023	341.2 (19.8)	-492.3 (-7.2)	-118.8(-1.1)	848.1 (48.9)	-578.3 (-17.4)
1984 – 2023	-49.6 (-1.2)	11.6 (0.2)	135.4 (2.3)	-118.5 (-1.5)	21.1 (1.5)

N.B.: PF = primary forest; SF = secondary forest; AL = agricultural land; FL = fallow land; BL/BU = bareland/built-up

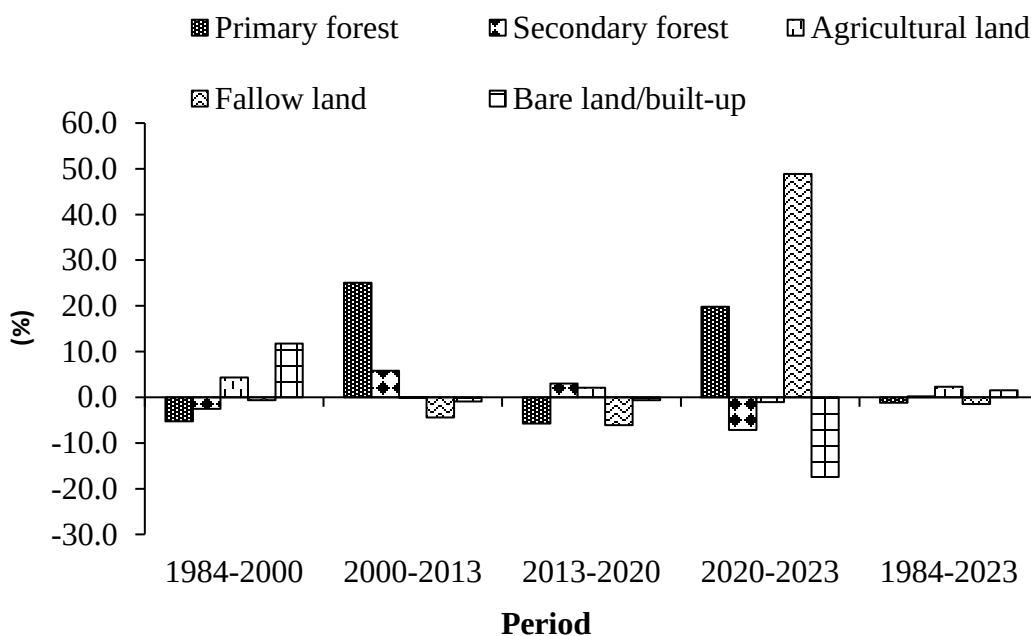


Figure 6: Annual rate of change in AFR between 1984 and 2023

Table 9 presents the confusion matrix, Table 10 presents the classification accuracy statistics generated, while Table 11 presents the summary of the accuracy statistics. User's accuracy is a more relevant measure of the classification's actual utility in the field. Secondary forest and bareland/built-up were found to be more reliable with 89% each of user accuracy. The commission error reflects the points, which are included in the category while they really do not belong to that category. The value was highest in the case of agricultural land. The omission error reflects the number of points, which were not included in the category while they really belong to that category. The omission error in case of secondary forest was highest (18%).

The overall classification accuracy was 85% with the average commission and omission errors being 4% and 15%, respectively. The average user accuracy and producer accuracy were 85% and 85%, respectively. The average specificity was 96%. The overall Kappa coefficient of 80% was obtained, which is rated as substantial.

Table 9: Confusion matrix for the LULC map

LULC	PF	SF	AL	FL	BL	Row Total
Primary forest	18	5	0	0	0	23
Secondary Forest	2	40	3	0	0	45
Agricultural Land	2	3	51	4	0	60
Fallow Land	0	1	6	40	2	49
Bareland/built-up	0	0	1	2	24	27
Column Total	22	49	61	46	26	204

N.B.: PF = primary forest; SF = secondary forest; AL = agricultural land; FL = fallow land; BL/BU = bareland/built-up

Table 10: Category-wise accuracy statistics for the LULC classification

LULC	Parameters					
	Sensitivity	Specificity	CE	OE	UA	PA
Primary forest	0.8182	0.9725	0.0275	0.1818	0.7826	0.8182
Secondary forest	0.8163	0.9677	0.0323	0.1837	0.8889	0.8163
Agricultural land	0.8361	0.9371	0.0629	0.1639	0.8500	0.8361
Fallow land	0.8696	0.9430	0.0570	0.1304	0.8163	0.8696
Bareland/built-up	0.9231	0.9831	0.0169	0.0769	0.8889	0.9231

N.B.: CE = Commission error; OE = Omission error; UA = Users accuracy; PA = Producers accuracy

Table 11: Accuracy statistics from the confusion matrix

Variables	Mean $\pm$ SD	
Average Omission error	0.15 $\pm$ 0.04	Overall classification accuracy = 85%
Average Commission error	0.04 $\pm$ 0.02	
Average User's accuracy	0.85 $\pm$ 0.05	
Average Producer's accuracy	0.85 $\pm$ 0.04	
Average Negative predictive power	0.21 $\pm$ 0.08	Overall Kappa coefficient 0.80 (substantial)
Average Sensitivity	0.85 $\pm$ 0.04	
Average Specificity	0.96 $\pm$ 0.02	

Table 12 shows the contingency matrices of changes within Aworo Forest Reserve between 1984 and 2023. The diagonal figures represent the static land-cover while other figures represent the matrices of change from one land-cover class to another. The frequency of land transformation from the contingency matrix showed that the study area experienced rapid LULC transformations within the period. Between 1984 and 2023, primary forest lost 19.9% to secondary forest, 13.5% to agricultural land, 15.1% to fallow land and 0.7% to bareland/built-up, having about 50.8% of the primary forest cover unchanged. Within the period, 16.1% of secondary forest was lost to primary forest, 23.2% to agricultural land, 31.6% to fallow land and 6.3% to bareland/built-up while 22.8% remained unchanged. However, agricultural land lost 14.6% to primary forest,

25.3% to secondary forest, 31.0% to fallow land and 3.8% to bareland/built-up while 25.3% of agricultural land remained unchanged. Fallow land lost about 8.1% to primary forest, 14.0% to secondary forest, 23.8% to agricultural land and 8.9% to bareland/built-up having 45.1% unchanged. Bareland/built-up lost 10.8%, 14.0%, 25.2% and 38.3% to primary forest, secondary forest, agricultural land and fallow land respectively while 11.7% of bareland/built-up remained unchanged. Details are presented in Table 12.

Table 12: LULC transition matrix (ha) for AFR between 1984 and 2023

LULC		LULC 2023 (ha)				
		PF (%)	SF (%)	AL (%)	FL (%)	BL (%)
LULC 1984 (ha)	PF	1215.37 (50.76)	478.57 (19.99)	322.99 (13.49)	361.26 (15.09)	16.14 (0.67)
	SF	924.82 (16.11)	1305.79 (22.75)	1333.40 (23.23)	1812.20 (31.57)	363.33 (6.33)
	AL	1639.76 (14.55)	2854.51 (25.33)	2855.71 (25.34)	3495.31 (31.02)	423.98 (3.76)
	FL	269.91 (8.09)	467.95 (14.02)	795.68 (23.84)	1503.83 (45.06)	299.87 (8.99)
	BL	232.19 (10.80)	301.75 (14.04)	541.33 (25.18)	822.49 (38.26)	251.89 (11.72)

N.B.: PF = primary forest; SF = secondary forest; AL = agricultural land; FL = fallow land; BL /BU= bareland/built-up

4.3 Drivers of Land-use and Land-cover changes in Aworo Forest Reserve

Figure 8 shows the contingency (transitional) matrix for possible transformations among the LULC within the study period in the study area. The main drivers of change

observed in the area were agricultural intensification and expansion; exploitation of forest for timber and fuel-wood, among other things.

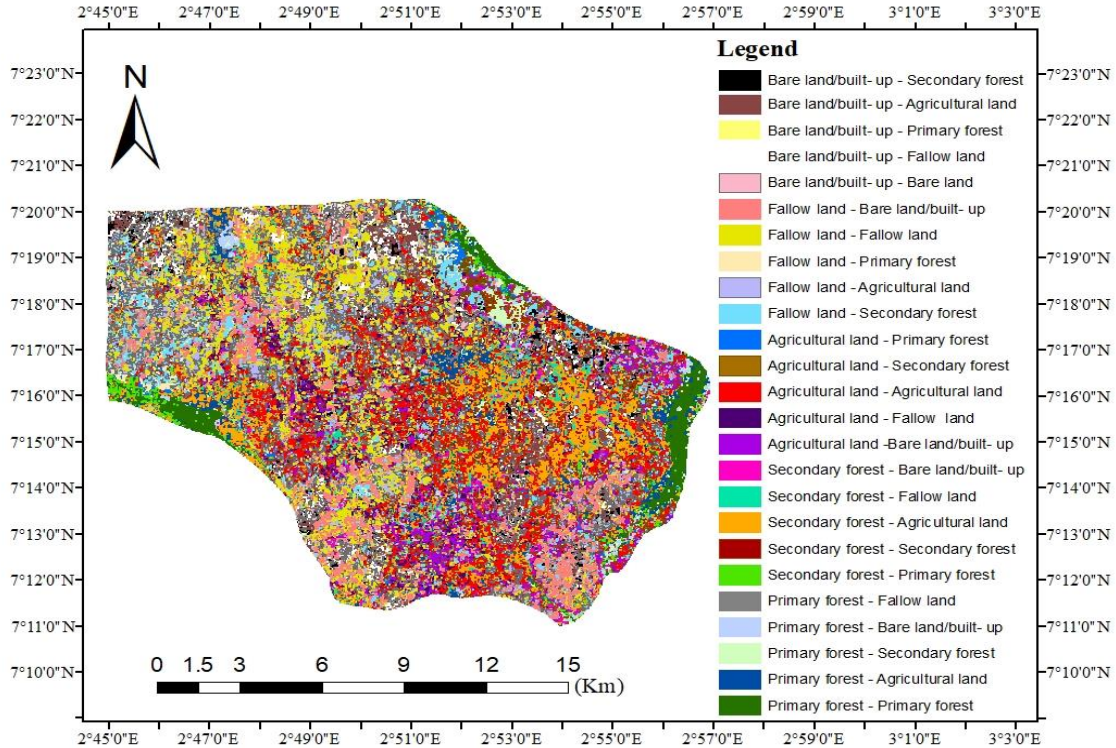


Figure 7: Transformation map for LULC in AFR between 1984 and 2023

4.4 Land-use and Land-cover Simulation of AFR between 2030 to 2120

The changes between 1984 and 2023 were used as the basis for prediction. Table 13 and Figure 9 present the simulation for the LULC in the area in the next 100 years, while figure 10 shows the map of selected years within the period of projections. The results revealed a decline in primary forest and fallow land to 2411 ha and 8060 ha,

respectively, while on the contrary, agricultural land, secondary forest and bareland/built-up, is expected to increase to about 24155 ha, 7013 ha and 4210 ha, respectively, in the year 2120, *ceteris paribus*.

Table 13: Land-use and land-cover Projection (ha) for the year 2030 to 2120

Year	PF (%)	SF (%)	AL (%)	FL (%)	BL (%)
2030	2056.5 (8.3)	5968.6 (23.9)	11967.6 (48.1)	2601.1 (10.4)	2312.7 (9.3)
2040	1560.1 (6.3)	6084.6 (24.4)	13321.7 (53.5)	1416.5 (5.7)	2523.5 (10.1)
2050	1063.8 (4.3)	6200.6 (24.9)	14675.8 (58.9)	231.9 (0.9)	2734.3 (10.9)
2060	567.4 (2.3)	6316.7 (25.4)	16029.9 (64.4)	-952.7 (-3.8)	2945.1 (11.8)
2070	71.1 (0.3)	6432.7 (25.8)	17384.1 (69.8)	-2137.3 (-8.6)	3155.9 (12.7)
2080	-425.3 (-1.7)	6548.7(26.3)	18738.2 (75.2)	-3321.9 (-13.3)	3366.8 (13.5)
2090	-425.3 (-3.7)	6664.7 (26.8)	20092.3 (80.7)	-4506.6 (-18.0)	3577.6 (14.4)
2100	-921.6 (-6.0)	6780.8 (27.2)	21446.4 (86.1)	-5691.2 (-22.9)	3788.4 (15.2)
2110	-1914.3 (-7.7)	6896.8 (27.6)	22800.5 (91.5)	-6875.8 (-27.6)	3999.2 (16.1)
2120	-2410.7 (-9.7)	7012.8 (28.2)	24154.7 (96.9)	-8060.4 (-32.4)	4210.0 (16.9)

N.B.: PF = primary forest; SF = secondary forest; AL = agricultural land; FL = fallow land; BL/BU = bareland/built-up

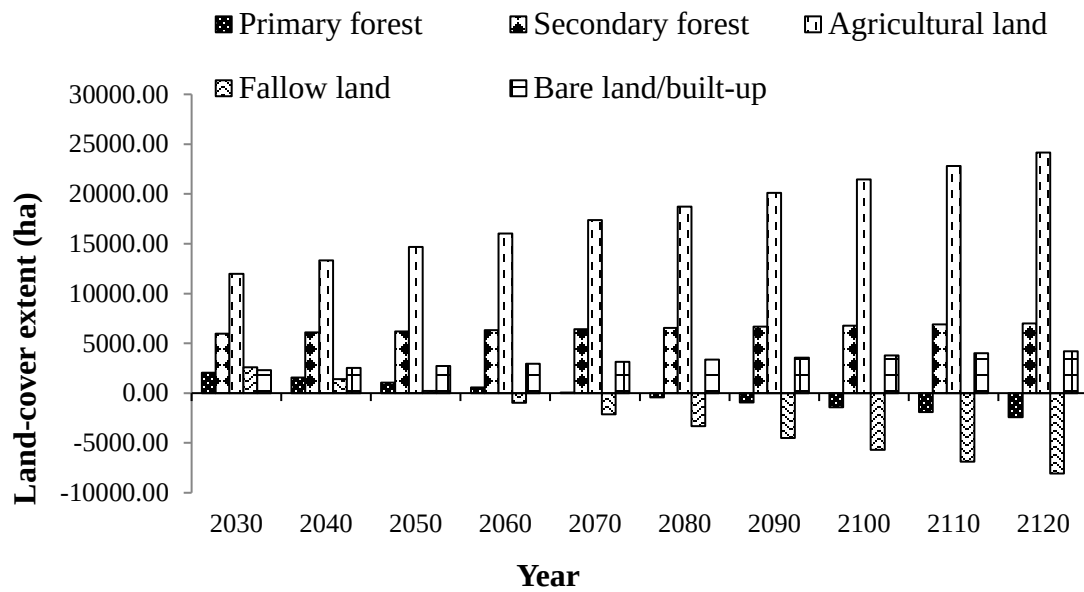


Figure 8: Land-use and land-cover Projection for the year 2030 to 2120

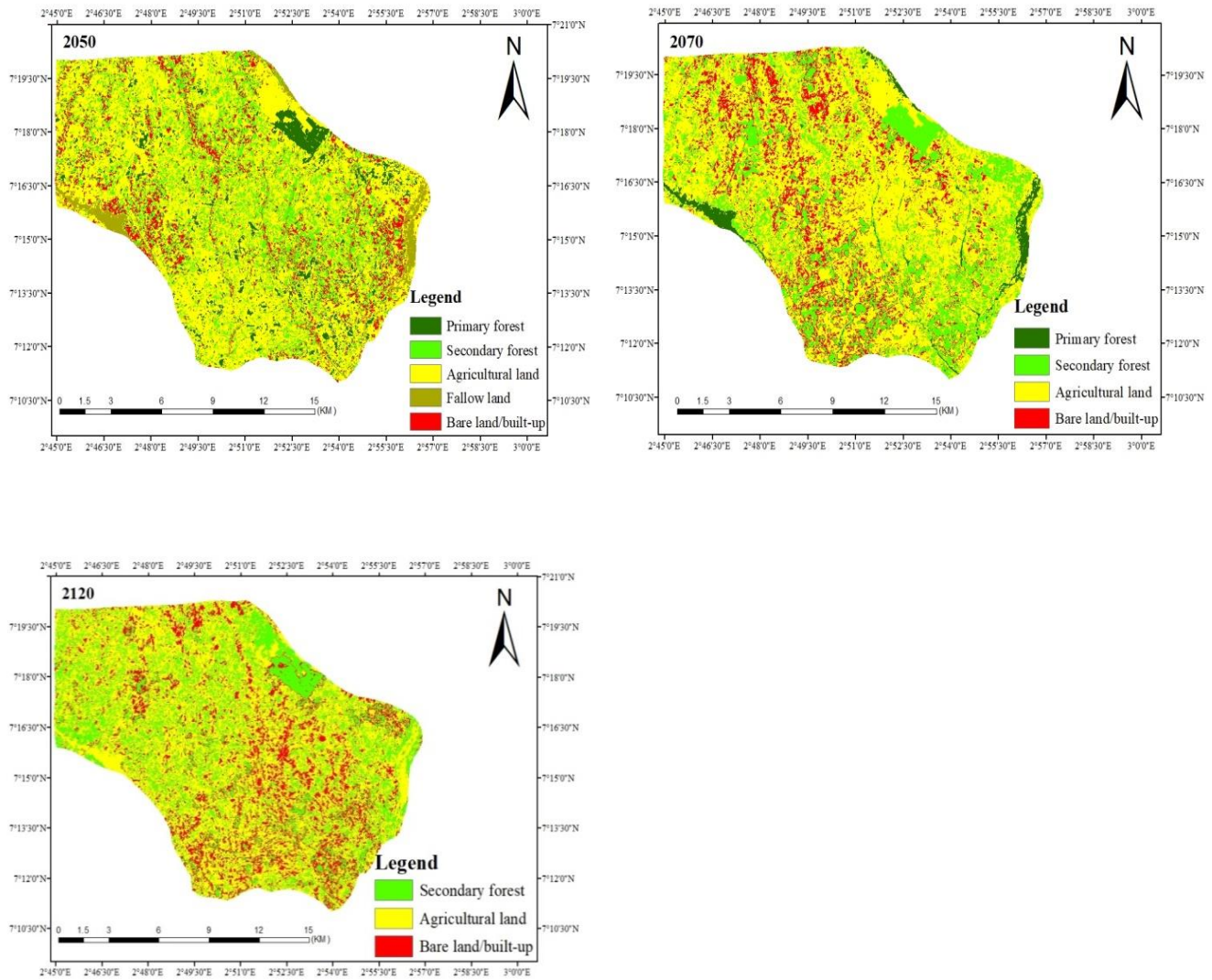


Figure 9: Future projection maps between 2050 and 2120

CHAPTER FIVE

5.0

DISCUSSION

The study showed that some level of forest losses occurred due to intensification of anthropogenic activities, which might have had negative impacts on flora and fauna biodiversity in the area. The trend in forest conversion can be attributed to land clearing for agricultural activities, illegal logging, and over-exploitation of forest resources. This is similar to the findings of Adeyemi and Ayinde (2022), who noted that over-exploitation and conversion to other land use are a result of significant human interference within reserved areas, which negatively impacted the luxuriance of natural vegetation. The demand for food arising from population increases may have influenced the dynamism of agricultural activities within the forest reserve, where increases in farming activities might have contributed to a decrease in forest-cover. Increase in fallow lands may have arisen from shifting cultivation/bush fallowing practices, in a bid to access virgin lands (in forested part) for agricultural purposes, while extending the forest depletion in the area. Moreover, extensive demand for virgin agricultural land and taungya system of farming is particularly largely responsible for the decrease in forest area, and if not controlled, will result in increased conversion of primary forest due to its perceived higher soil fertility.

The dynamism in land use observed within the area may have also arisen due to high level of poverty resulting in increased demand for forest resources in forms of

livelihood, such as high demand for farmland, exploitation of timber and non-timber forest products. This is in consonance with the finding of Munthali *et al.* (2019) that, majority of the local communities surrounding forests Dedza are characterized by poverty and lack of alternative to livelihoods. Therefore, the decline in forest-cover can be attributed to poverty and rapid population growth, which creates enormous pressure, competition and over-dependence on forests, water and land, causing unsustainable exploitations of these resources with implications for biodiversity, habitats, ecosystem services, and for people's livelihoods. This is similar to the findings of Lei *et al.* (2007), who opined that the rapid increase in human population and strive for growth in the standard of living has put great pressure on natural resources in the forests. Although it was revealed by Oduntan *et al.* (2013), that recent effort and enforcement has been put in place by the management of the reserve in evacuating all illegal occupant of forest land, which will help check high degree of land-use conversion in the area, stricter measures may be needed in forestalling the projected loss in forest cover.

Generally, anthropogenic activities were observed to be the major disturbance of forest-cover in the area, and the influence of such activities has intensified remarkably in the area during the last few decades. This is established by the findings of Oduntan *et al.* (2013), who observed that pressures from logging and harvesting of non-timber forest product were becoming severe in most natural forests. Hunting, grazing and cross-boundary influence were also observed as drivers of forest cover loss within the area.

As stated by Rahdary *et al.* (2008), LULC change is dynamic, widespread and accelerating process, mainly driven by natural phenomena and anthropogenic activities, and it is evident that the conversion and intensification of land-use by humans have caused huge changes in the balance of natural ecosystems, particularly in Aworo Forest Reserve.

CHAPTER SIX

6.0 CONCLUSION AND RECOMMENDATIONS

6.1 Conclusion

The study has revealed that changes in the LULC pattern from 1984 to 2023 within the study area are closely defined by the ever-increasing human needs, which has resulted to forest losses and detrimental land-use dynamics. This study demonstrates the ability of GIS and remote sensing in capturing spatial-temporal data. Attempt was made to capture as accurate as possible five land-use land-cover classes as they change through time. The observed forest-cover losses were traced to increasing agricultural land and proliferation of rural settlements. Thus, the extent of the agricultural land had increased between 1984 and 2023. These changes have come at the expense of the forest-cover, and may have adverse environmental implications, if the trend continues.

This study successfully shows that the change detection techniques can be applied to land-use/land-cover change using remote sensing data. Therefore, the use of remotely sensed data and robust image-processing technique offer accurate, timely and detailed information for detecting and monitoring changes originating from land-uses overtime.

6.2 Recommendations

It is important that local level natural resource management programme be put in place to reduce the harmful impact of land-use/land-cover change resulting from the

degradation of natural forest. Special emphasis should be on enhancing the quality and quantity of the existing forest cover. This study has been carried out in Aworo Forest reserve, using these modern technologies in conjunction with field observation, which showed land-cover conversion and modification. The problems associated with the land-use/land-cover changes discussed are linked to increasing demand for agricultural land and illegal logging. The general trend observed by the present study is a decrease in primary forests and fallow land and a corresponding increase in agricultural land, secondary forest and bareland/built-up. The implication is that the present activities may lead to more land degradation. Thus, this trend should be checked. Land-use/land-cover mapping and detection of changes shown by this study will serve as a basis to understand the patterns and possible causes and consequences of land-use/land-cover changes in the area.

The findings of this study highlights the need for a comprehensive assessment of human activities in Aworo Forest Reserve and the adaptation of sustainable forest management practices such as close supervision of the forest reserve and restoration of already degraded forests.

Furthermore, the developed spatial maps would serve as an efficient tool for spatial analysis and spatial modelling functions to gain insight into a prevailing situation in the forest reserve. It is recommended for use in formulating meaningful policies so as to achieve a sustainable forest management in the area.

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